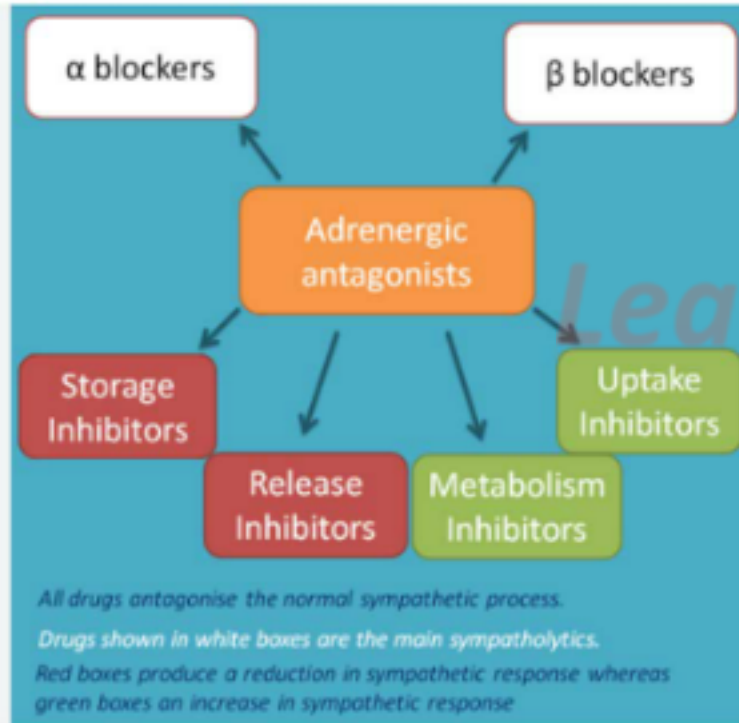


ADRENERGIC ANTAGONISTS



🎯 Alpha adrenergic blockers

🎯 Beta adrenergic blockers

Adrenergic Antagonists

- Adrenergic is a term that refers to boosting or blocking the effects of the neurotransmitter epinephrine and norepinephrine in the body.
- The adrenergic receptors are located in the heart, lungs, kidneys, and gastrointestinal system.
- Before learning about agonists and antagonists, it's important to understand affinity and intrinsic activity.

Affinity :

- The ability of a drug to bind with a receptor.

Intrinsic activity :

- The ability of a drug to produce conformational changes in the receptor and produce biological response.

Agonist :

- A chemical that binds to a receptor and activates the receptor to produce biological response .

Properties :

- It has affinity property and intrinsic activity.

Agonist :-

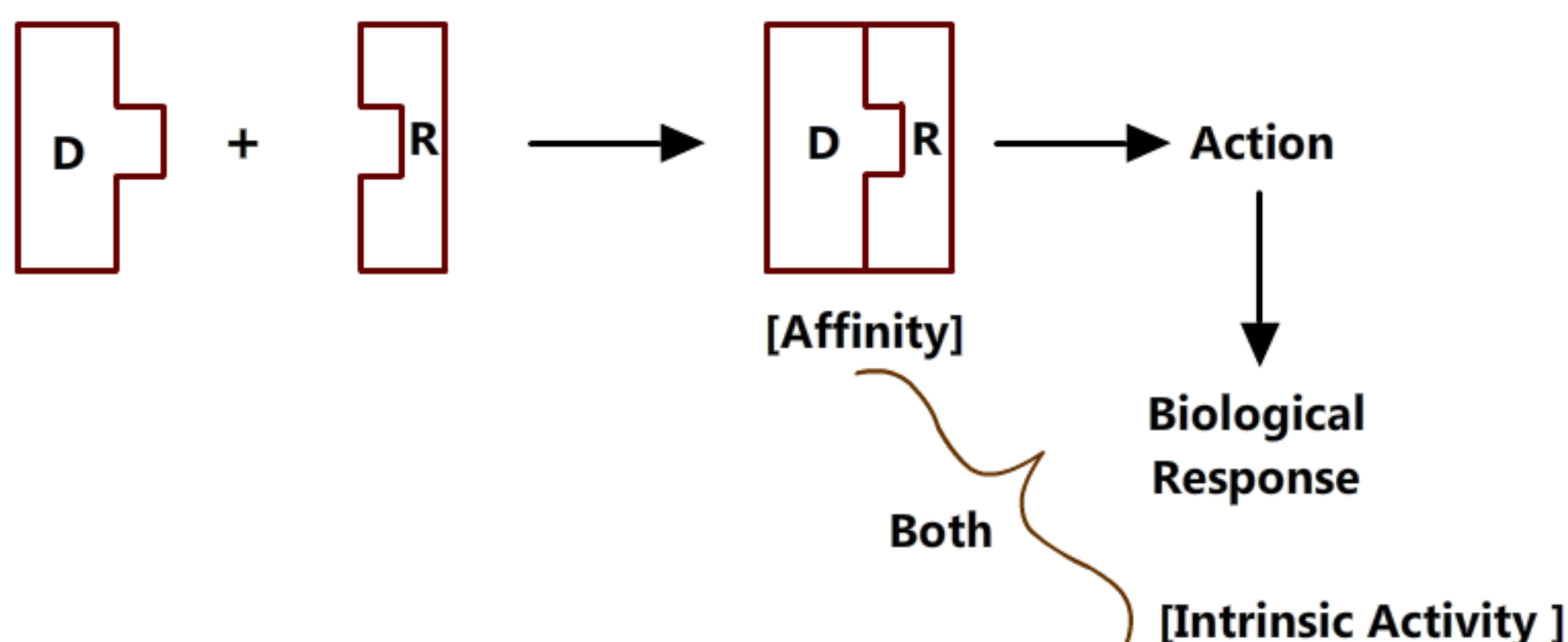


Fig 1: Agonist activity

Antagonist :

- A chemical that binds to a receptor but inhibits the activity of the receptor or blocks the action.
- It has only affinity property.

Antagonist :-

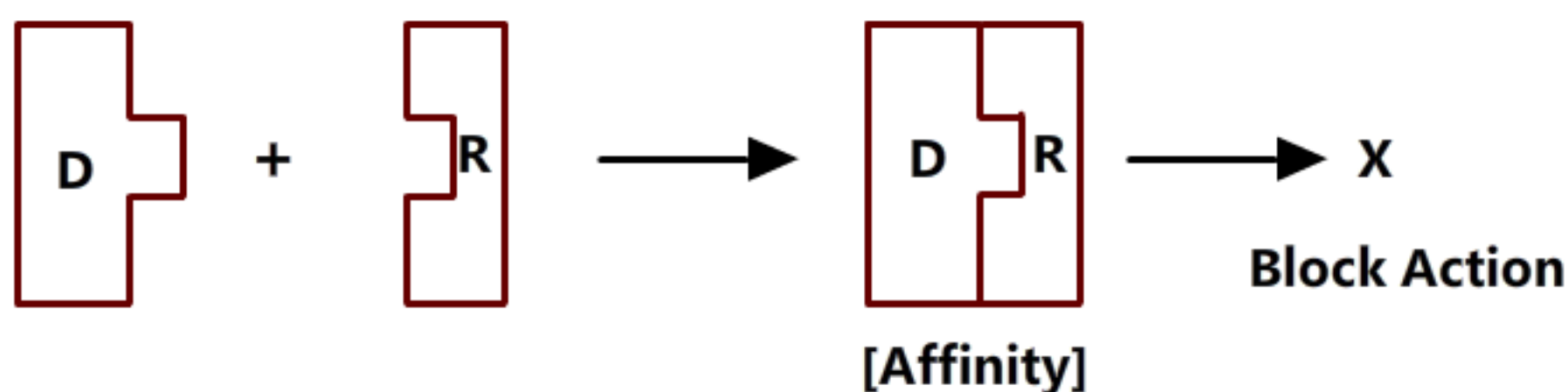


Fig 2: Antagonist activity

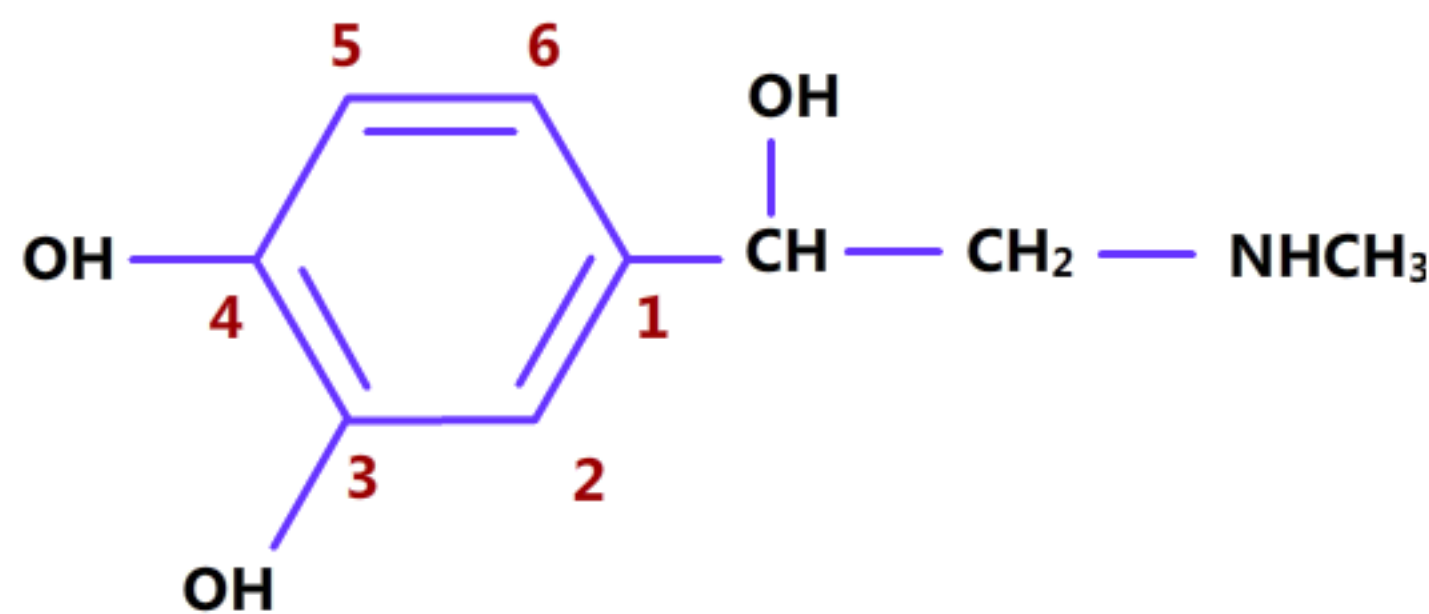
Adrenergic antagonist :

- Adrenergic antagonist is also called Sympatholytics and inhibition of a sympathetic activity is antagonism.
- Adrenergic antagonists are drugs that block the effect of adrenaline, noradrenaline, and other catecholamines, which control autonomic flow and various processes of the central nervous system.
- Adrenergic antagonists bind to adrenergic receptors to elicit its blocking activity.
- The first group of receptors are alpha adrenergic receptors and the second group of receptors are beta adrenergic receptors.
- There are two types:
 - Alpha adrenergic receptors: alpha 1 and alpha 2.
 - Beta adrenergic receptors are divided into beta 1, beta 2 and beta 3.
- Adrenergic drugs which stimulate (agonist) and inhibit (antagonist) adrenergic receptors and it mediates the action of adrenaline and noradrenaline.

Adrenaline :

- Adrenaline is a hormone which is also called Epinephrine and it is involved in the fight or flight response, in our body it releases in emergency situations.

- For example: if a bear comes before us, we need to prepare to run from that bear, so adrenaline is released.
- It increases the heart rate, force of contraction and cardiac output.
- Adrenaline guides for bronchodilation(make breathing easier by relaxing the muscles in the lungs and widening the airways) and it mediates the liver to release glucose.
- It helps in vasodilation of muscles(the widening of blood vessels as a result of the relaxation of the blood vessel's muscular walls) which is secreted from the adrenal medulla.

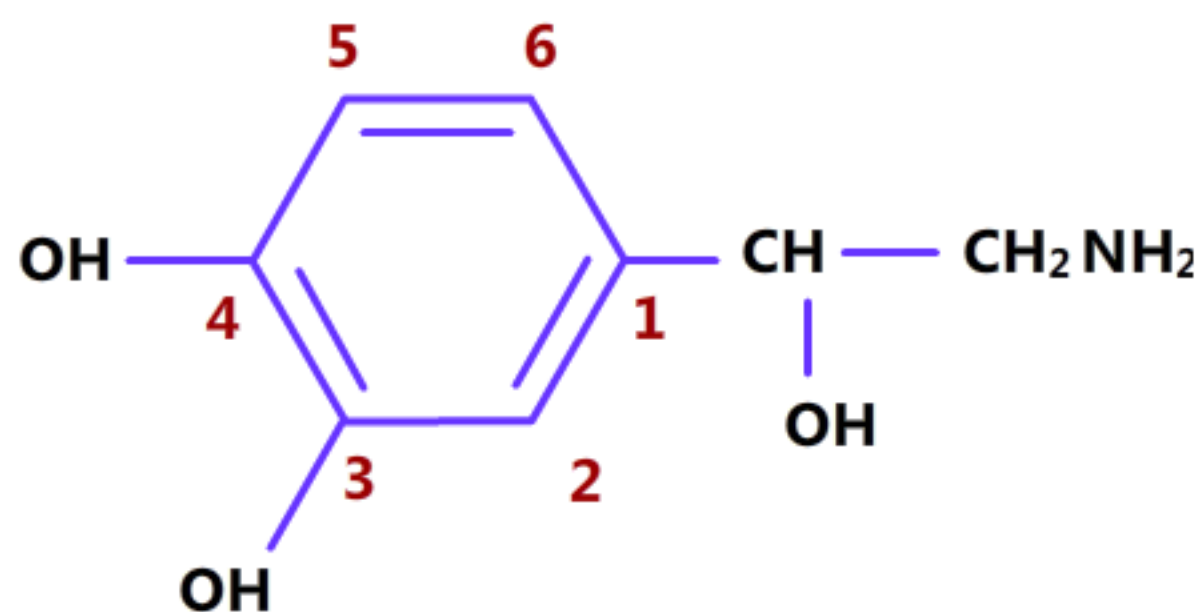


Adrenaline
Fig 3: The structure of Adrenaline

- IUPAC Name:** R(1-(3,4-dihydroxy phenyl)-2-methyl amino ethanol.
- The presence of methyl ring in the 1st position and it has benzene rings because they are called catecholamines(class of aromatic amines) and this benzene ring attaches to the methyl amino group and the difference is adrenaline is methylated.

Noradrenaline :

- Noradrenaline is a neurotransmitter and it is also called Norepinephrine.
- It is produced by adrenergic neurons.
- It has strong vasoconstriction properties.
- It elevates the blood pressure.
- It is given through an intravenous route due to its poor oral bioavailability.



Norepinephrine/Noradrenaline

Fig 4: Structure of nor-epinephrine or nor-adrenaline

- **IUPAC Name:** (R)-2-Amino-1-(3,4-dihydroxyphenyl)ethanol.
- Benzene ring attached to the amino group with 2 carbon chain in the 1st position and hydroxyl groups attached to the 3rd and 4th position are also called catecholamines (presence of benzene rings).
- The difference for the adrenaline and noradrenaline was the adrenaline which has methyl group but the non adrenaline does not have the methyl group on amine.

Mechanism of action of Adrenergic antagonist :

- Adrenergic antagonists are also called as sympatholytics which bind to the adrenergic receptors.
- Adrenergic receptors are G protein coupled receptors that are going to elevate the secondary messenger system in our body and produce physiological effects.
- The neurotransmitter releases and binds to the alpha receptor and beta receptors and the activity starts and produces a biological action.
- When the neurotransmitter is released and binds to alpha receptors, the adrenergic antagonist drugs are used to block the action and result in a decrease in heart rate called vasodilation.
- The mechanism of action refers to the blockage or inhibition of alpha and beta receptors.

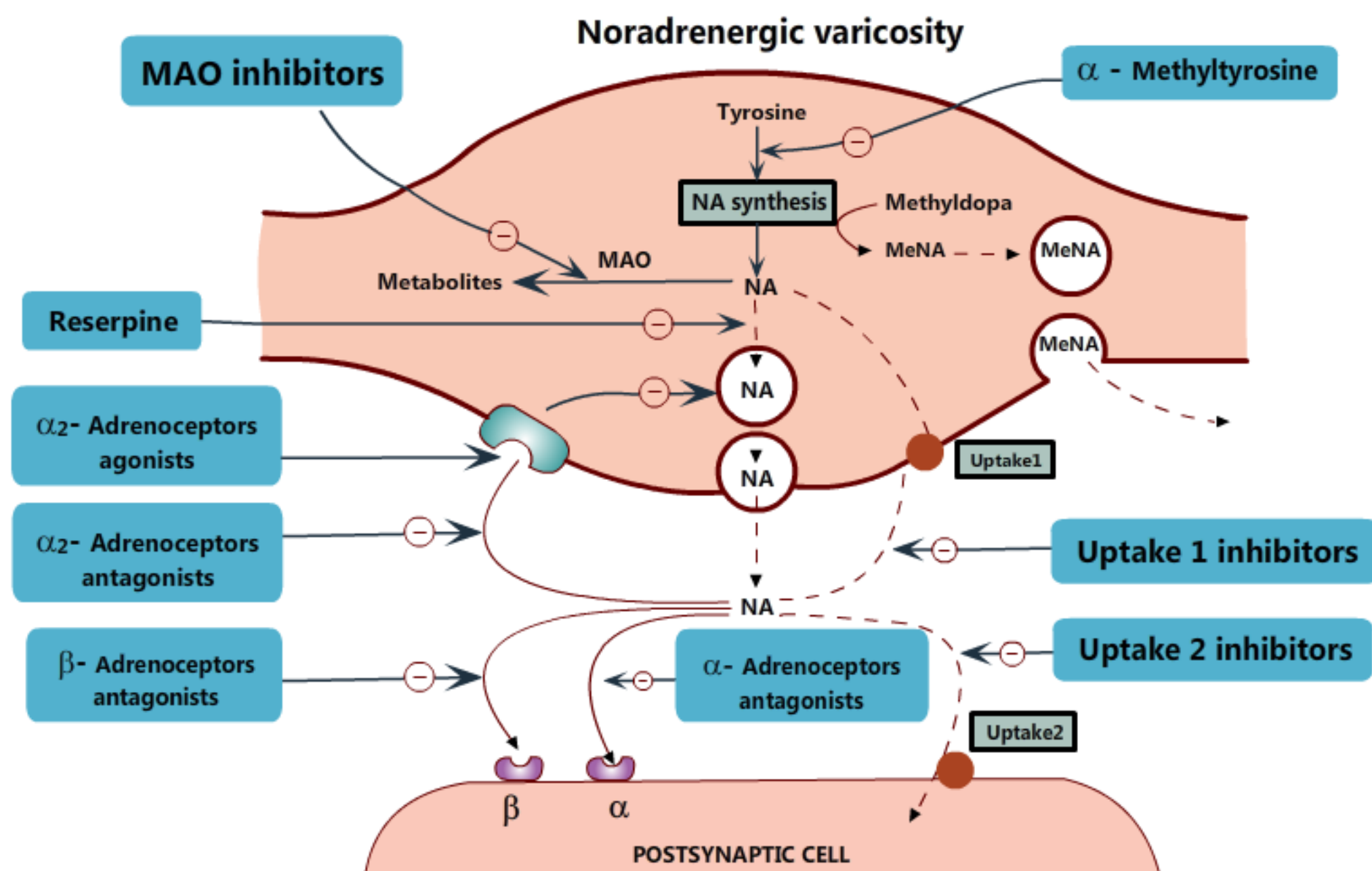


Fig 5: Mechanism of action of Adrenergic antagonist

Classification :

- Adrenergic antagonists are classified into alpha adrenergic blockers and beta adrenergic blockers.

Alpha adrenergic blocking drugs :

- Alpha 1 selective: Prazosin, Terazosin, Doxazosin, Alfuzosin, Tamsulosin, Sildenafil.
- Alpha 2 selective: Yohimbine.
- Non Selective: Tolazoline, Phenoxybenzamine, Ergotamine, Ergotamine, Phentolamine, Chlorpromazine, Dihydroergotamine, Dihydroergotamine.

Beta adrenergic blocking drugs :

- Non- selective(beta 1 and beta 2) antagonists: propranolol, sotalol, Timolol, Pindolol, Labetalol, Carvedilol.
- Selective (Beta 1) antagonists: metoprolol, Atenolol, Acebutolol, Bisoprolol, Esmolol, betaxolol, celiprolol, Nebivolol.

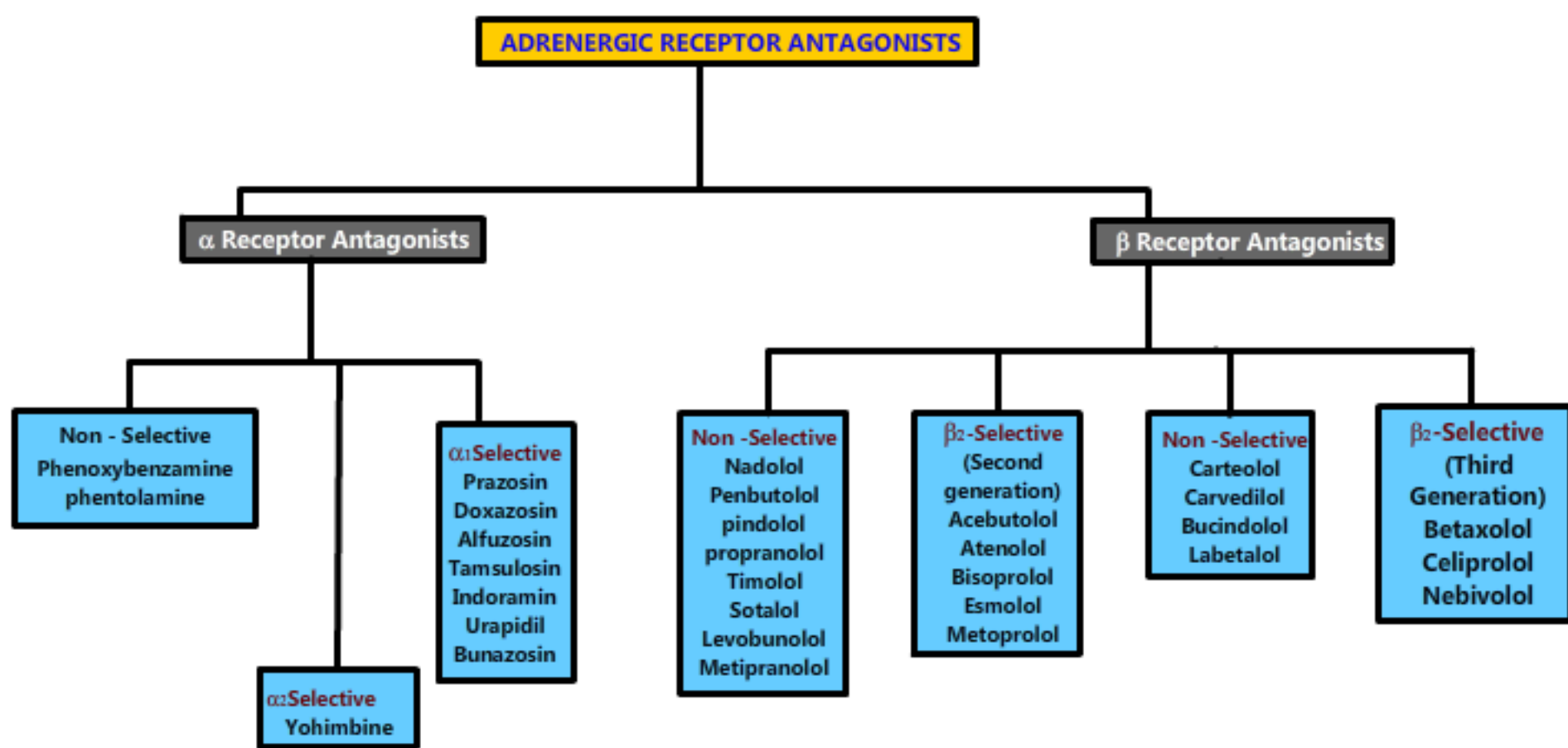


Fig 6: Mechanism of action of Adrenergic antagonist

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Alpha adrenergic blockers

- These drugs inhibit adrenergic responses mediated through the alpha adrenergic receptors without affecting those mediated through beta receptors.
- Alpha adrenergic blockers block the action of adrenergic agents on alpha receptors.
- These drugs act on either alpha 1 or alpha 2 receptors or non selective on the receptor.
- Alpha adrenergic blockers are used to treat high blood pressure.
- Alpha adrenergic blockers consist of selective and non selective blockers.
- Selective blockers are neurotransmitters that only attach to one of two receptors: the alpha 1 or alpha 2 receptor.
- Neurotransmitter non-selective blockers bind to both alpha 1 and alpha 2 receptors
- The drugs are studied below: Tolazoline, Phentolamine, Phenoxybenzamine, prazosin, dihydroergotamine, methysergide.

Tolazoline :



- Tolazoline is made by combining toluene and imidazoline, and it is distinguished by its structural representation.
- Below is a detailed representation of Tolazoline.

Tolazoline:-Totuene + Imidazoline

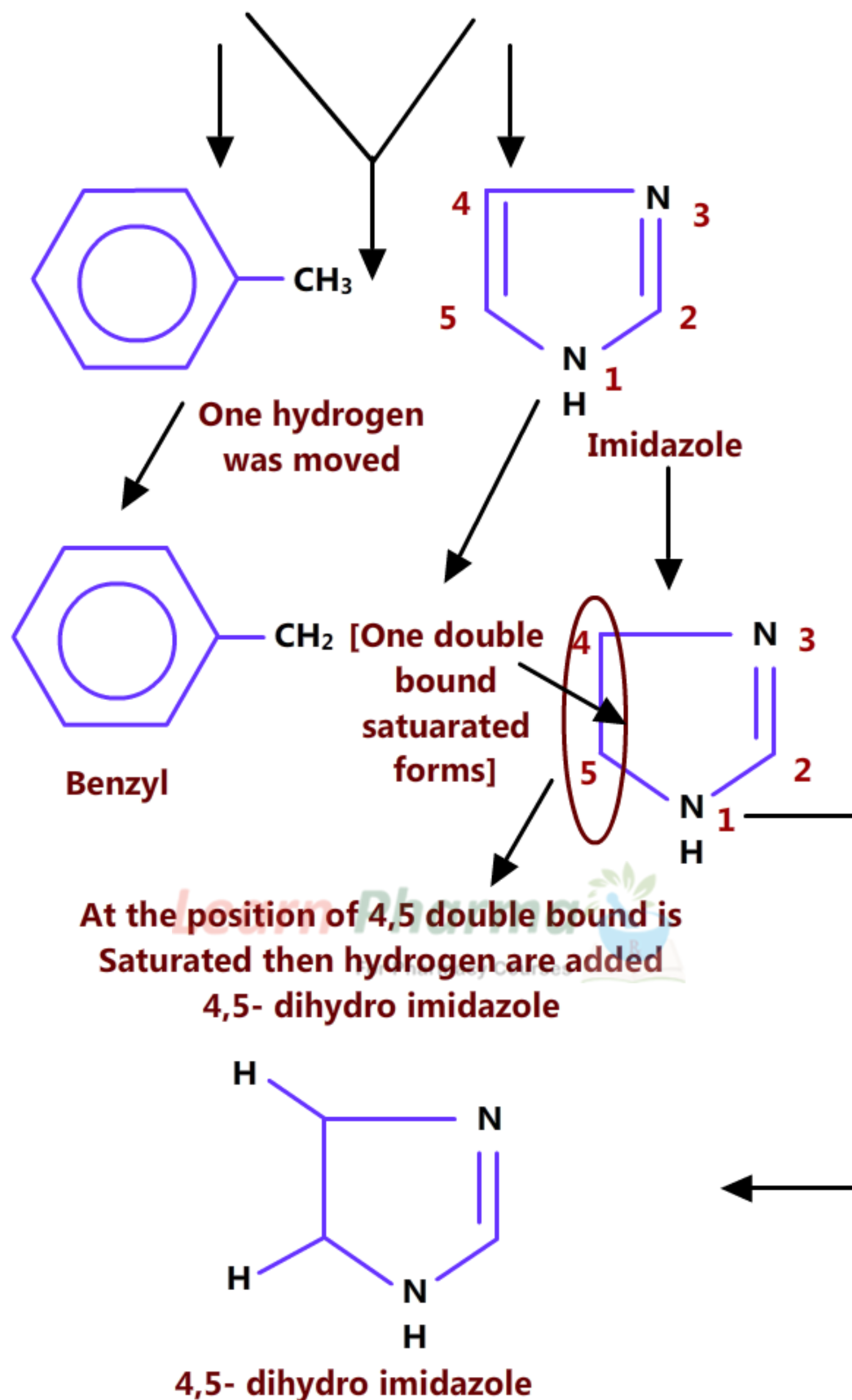


Fig 7: The detailed representation of Tolazoline

Structure of Tolazoline :

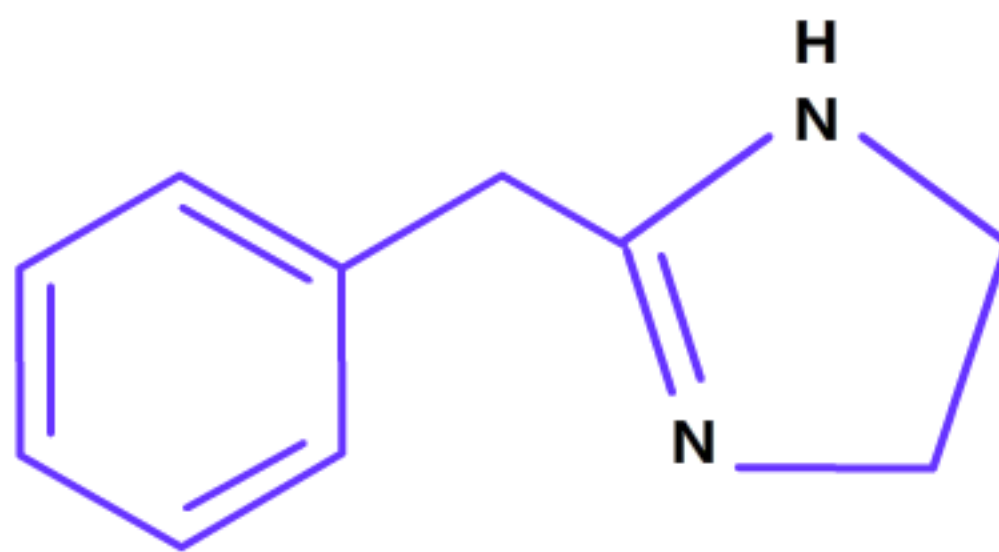


Fig 8: The structure of Tolazoline

- **IUPAC name:** 2-Benzyl-4,5 dihydro-1H-imidazole / 2-benzyl-2-imidazoline
- **Molecular Formula:** $C_{10}H_{12}N_2$

Properties :

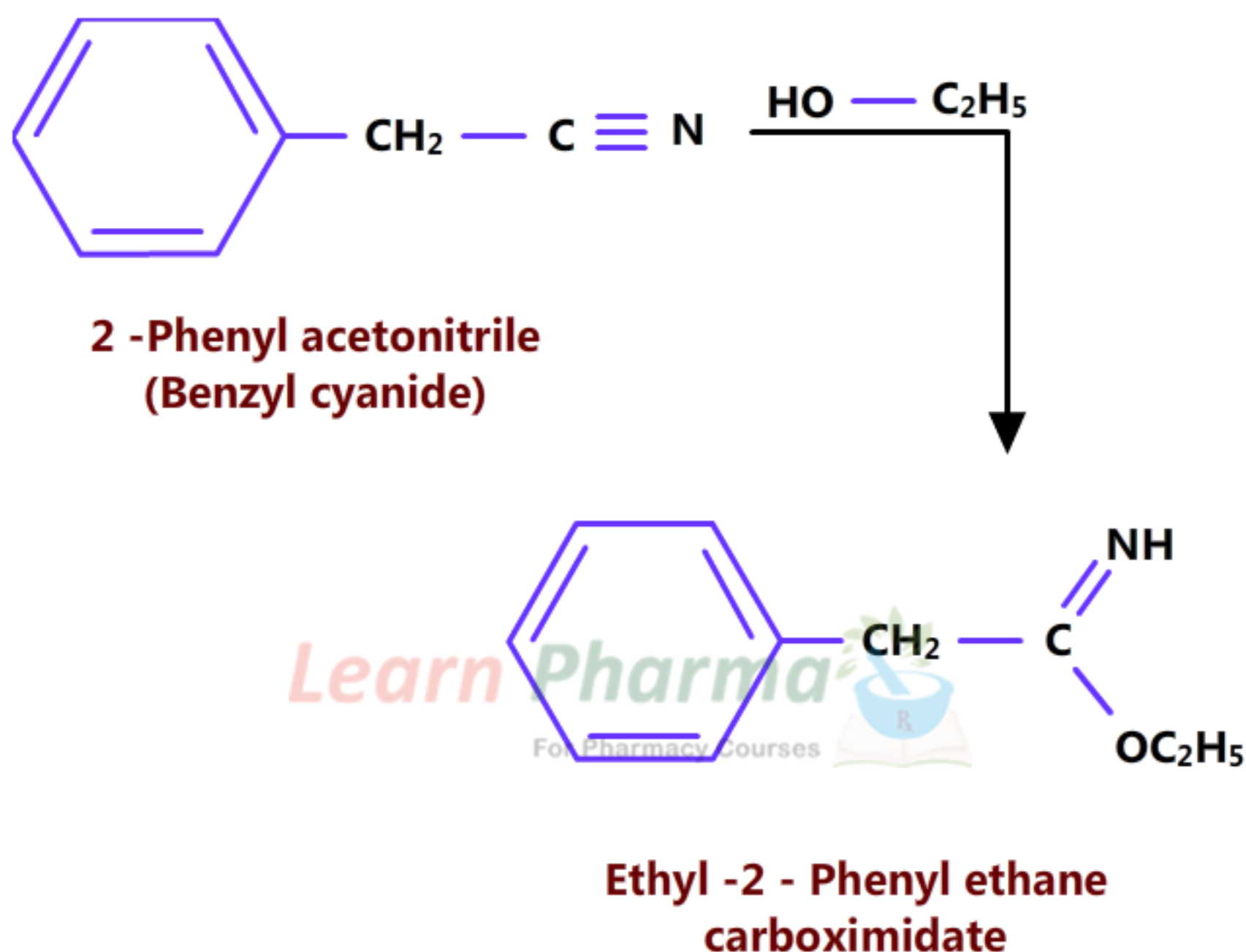
- It is white or creamy white microcrystalline powder
- It has odourless
- It has a slightly bitter taste
- It is very soluble in water
- It is freely soluble in alcohol.
- **Storage:** it is stored in a well closed light resistant container at a cool place away from direct sunlight.



Synthesis of Tolazoline :

- Tolazoline is a directly acting non selective adrenergic antagonist at alpha-adrenergic receptor and used to treat pulmonary hypertension occurring in neonates and also used in the treatment of acute or chronic vascular hypertension regardless of pharmacological mechanism.
- The synthesis of tolazoline begins with the starting material, 2-phenyl acetonitrile is treated with ethanol, leading to a nucleophilic addition reaction.
- During this reaction, ethanol being an acid, readily produce H^+ ion which is added to the nitrogen atom of the nitrile group.
- Meanwhile the ethoxy group ($-OC_2H_5$) of the ethanol attacks on the electrophilic carbon of the nitrile group forming ethyl 2-phenyl ethane carboximidate as an intermediate product.
- In the second step, ethyl 2-phenyl ethane carboximidate is treated with 1,2-diamino ethylene, leading to the cyclization.

- During this type of reaction, the hydrogen atom of the imine group reacts with one of the amino group of 1,2-diamino ethylene and subsequent losses of ammonia.
- Meanwhile another amino group of the same compound attacks on the carbon atom of the imino group.
- This is followed by the ethoxy group ($-\text{OC}_2\text{H}_5$) of the intermediate and the hydrogen atom of amine combinedly eliminated as ethanol, yielding the final desired product tolazoline.



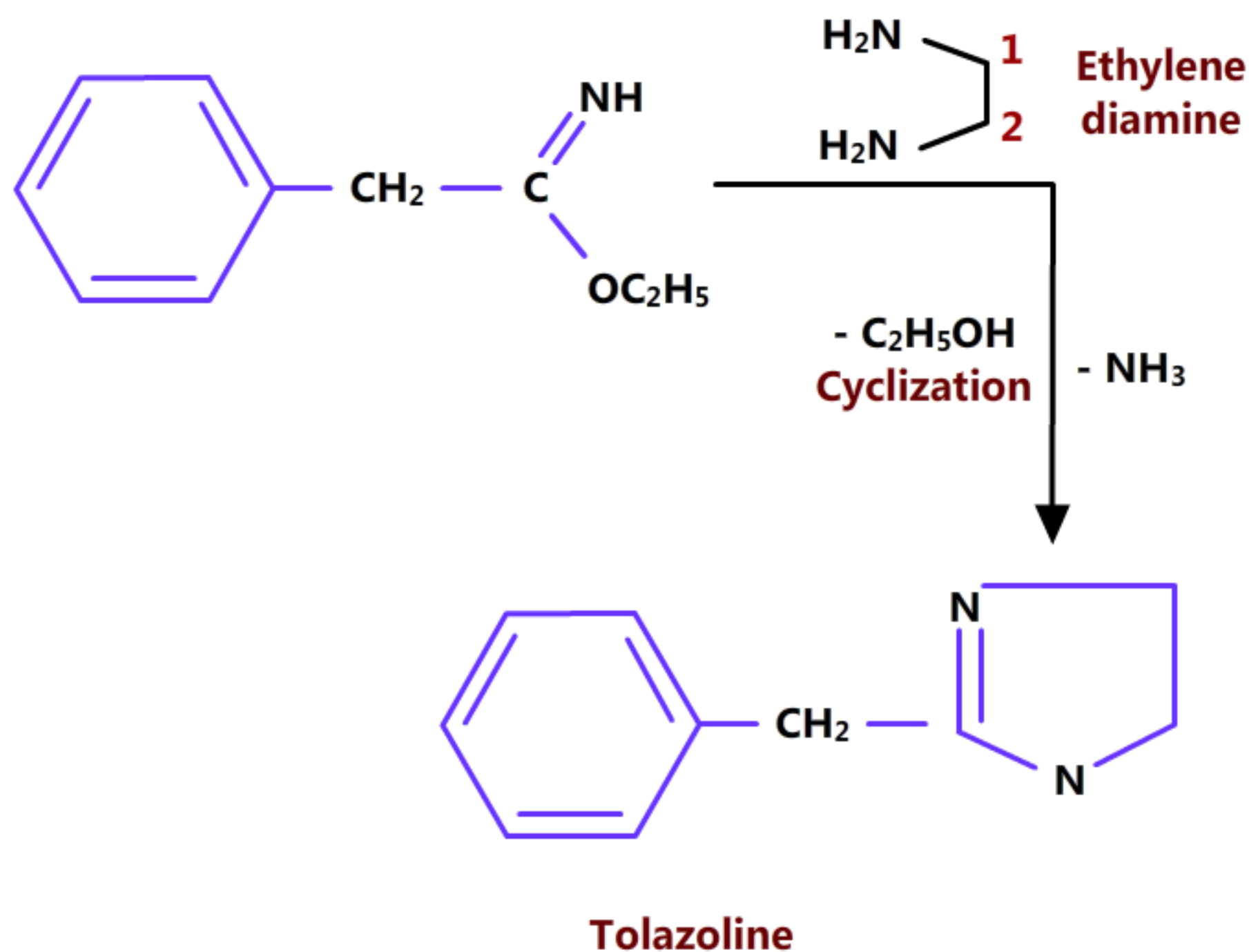


Fig 9: Synthesis of Tolazoline

Mechanism of Action :

- Alpha 1 receptors are present on the vascular smooth muscle, and when norepinephrine and epinephrine can act on the alpha 1 receptor they prolong vasoconstriction then increases blood pressure and increases in the total peripheral resistance, when alpha blockers are given it act on the alpha 1 receptors and then blocks and consists of vasodilation and decrease in the blood pressure and decrease in the TPR.
- Alpha 1 receptors are found on the smooth muscle of the vascular system, and when norepinephrine and epinephrine act on them, they prolong vasoconstriction, which raises blood pressure and the arteries, resulting in an increase in the TPR. When alpha blockers are given, they act on the alpha 1 receptors and block them, resulting in vasodilation, lower blood pressure, and a decrease in the TPR.

Structure activity relationship :

- The structure of 2,6 substitutions of the phenyl and imidazoline rings shows highest activity of the receptor binding.
- Where the electronic configuration takes place in the reaction undergoes in the H₂ receptor actions where the action of alpha receptor activity was not influenced.

- If the substitution takes place in 3,4 and 5th positions, the phenyl ring undergoes the activity of H₂ receptor sites while the alpha receptor activity is maintained.

Uses :

- It is used in the spasm of peripheral blood vessels.
- It is used in the area of the body if we feel numb and cool.
- It is used in the inflammatory process due to a clot in the legs.
- It is used in chronic arm or leg pain during injury, surgery, stroke and heart attack.

Phentolamine :

- Phentolamine is an alpha adrenergic antagonist and it is an imidazole derivative.
- It belongs to a non-selective alpha adrenergic blocker.
- This is used in the treatment of hypertensive conditions and used as vasodilators same as tolazoline.

Structure :

- The structure is composed of a phenyl ring attached with the OH group in 1st position and the amino ring attached to the 3rd position.
- To that amino tolyl group was attached and also imidazole was attached, which is a 5 membered carbon ring with the presence of nitrogen in the position of 1st and 3rd.



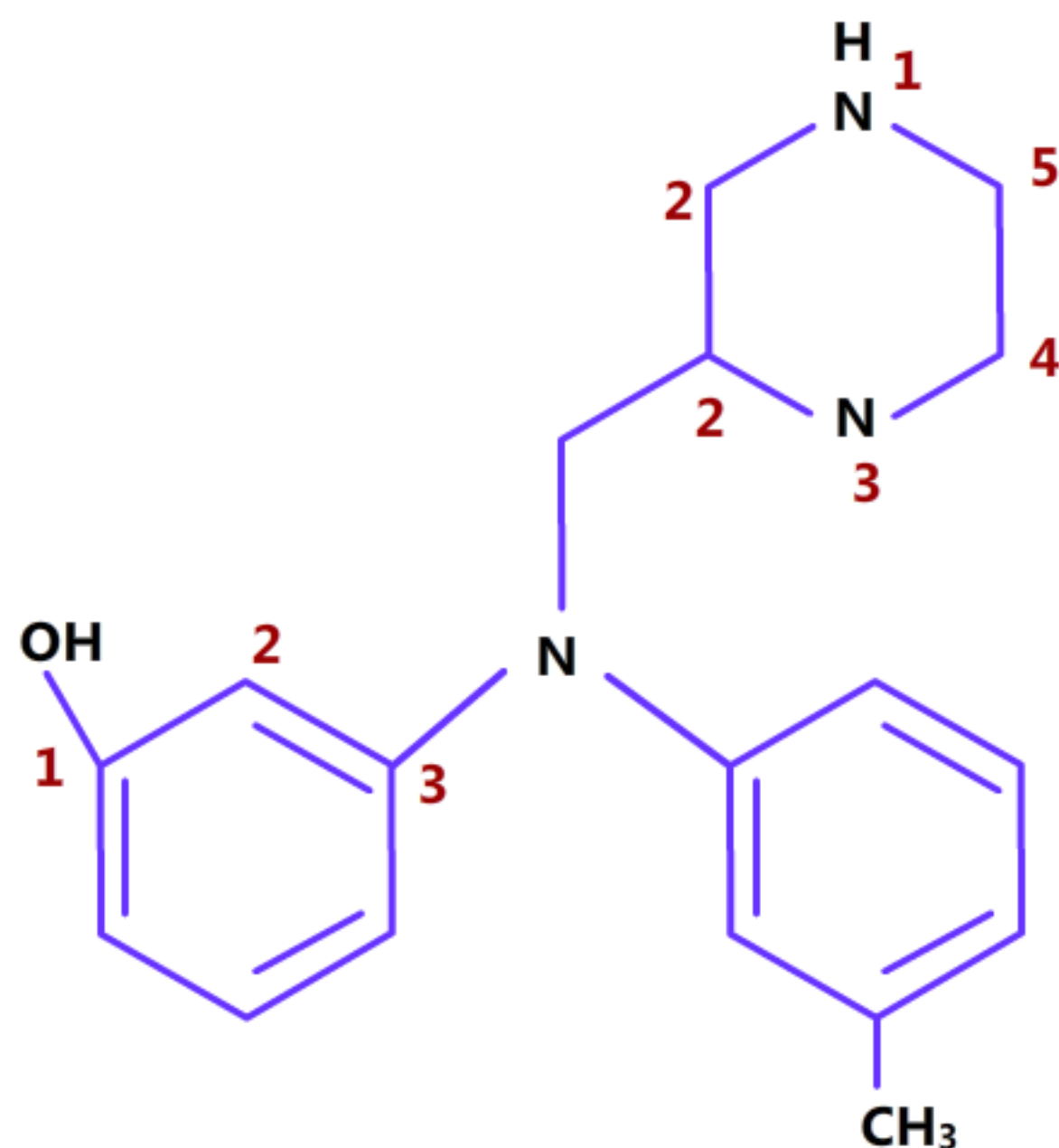


Fig 10: The structure of Phentolamine

- **IUPAC Name:** 3-[(4,5 dihydro-1H-Imidazole-2-yl-methyl)p-tolyl amino]Phenol
- **Molecular formula:** $C_{17}H_{19}N_3O$

Mechanism of action :

- It blocks both alpha 1 and alpha 2 receptors and it produces vasodilation and lowers blood pressure.
- It leads to relaxation of muscles and widening of blood vessels and widening of blood vessels in decrease in blood pressure.
- Phentolamine also stimulates beta adrenergic receptors and produces a positive inotropic and chronotropic effect on the heart which results in an increase in the cardiac output.

Structural Activity Relationship :

- Molecules with 2,6-substitutions, which assume an orientation where the phenyl and imidazoline rings are in different planes, are having the highest activity.
- Electronic effects have only influence on the actions at H_2 ; action on alpha receptors are not influenced.
- Substitutions at 3, 4 or 5 positions of the phenyl ring preclude potent activity at H_2 receptor sites while the alpha receptor activity is maintained.

Synthesis :

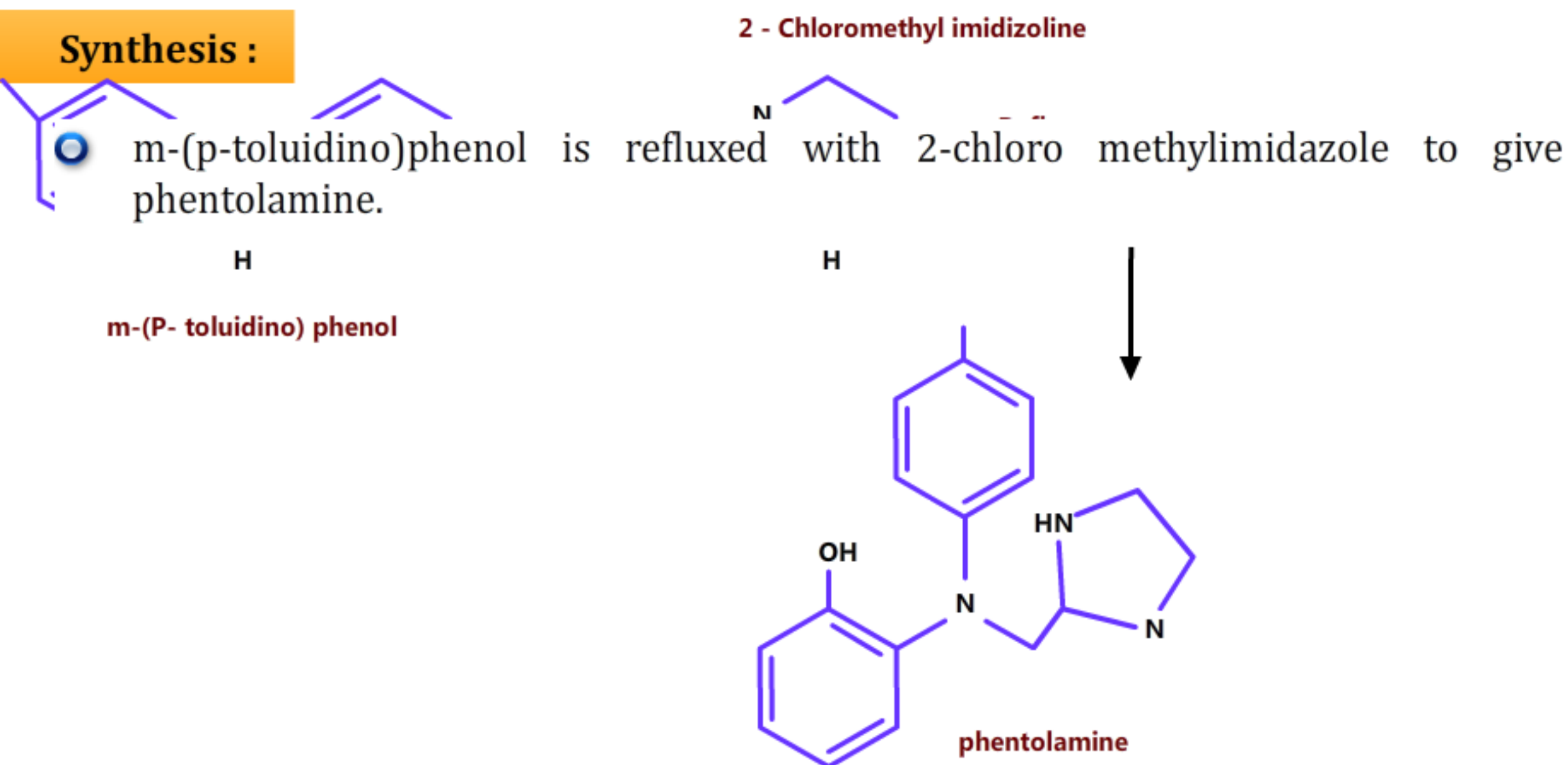


Fig 11: Synthesis of Phentolamine



Uses :

- Reversing numbness after dental procedures.
- For the prevention of events of high blood pressure.
- Treatment of accidental injection of certain drugs under the skin.
- It is used for the diagnosis and treatment of pheochromocytoma.
- It is used to treat Raynaud's syndrome.

Physicochemical properties :

- Molecular weight : 281.35 g/mol
- Physical appearance : Present in crystalline solid form
- Melting point : 174.5 °C
- Solubility : 4.59 mg/L in water
- Water partition coefficient : 3.3

- Presence of rings : Benzene and imidazoline rings are present

Phenoxybenzamine :

- Phenoxybenzamine is a non selective-alpha adrenergic blocker and it blocks alpha 1 receptor and it is used in the treatment of Tachycardia Arrhythmia and reduces the tumor of adrenal medulla and reduces blood pressure.

Structure :

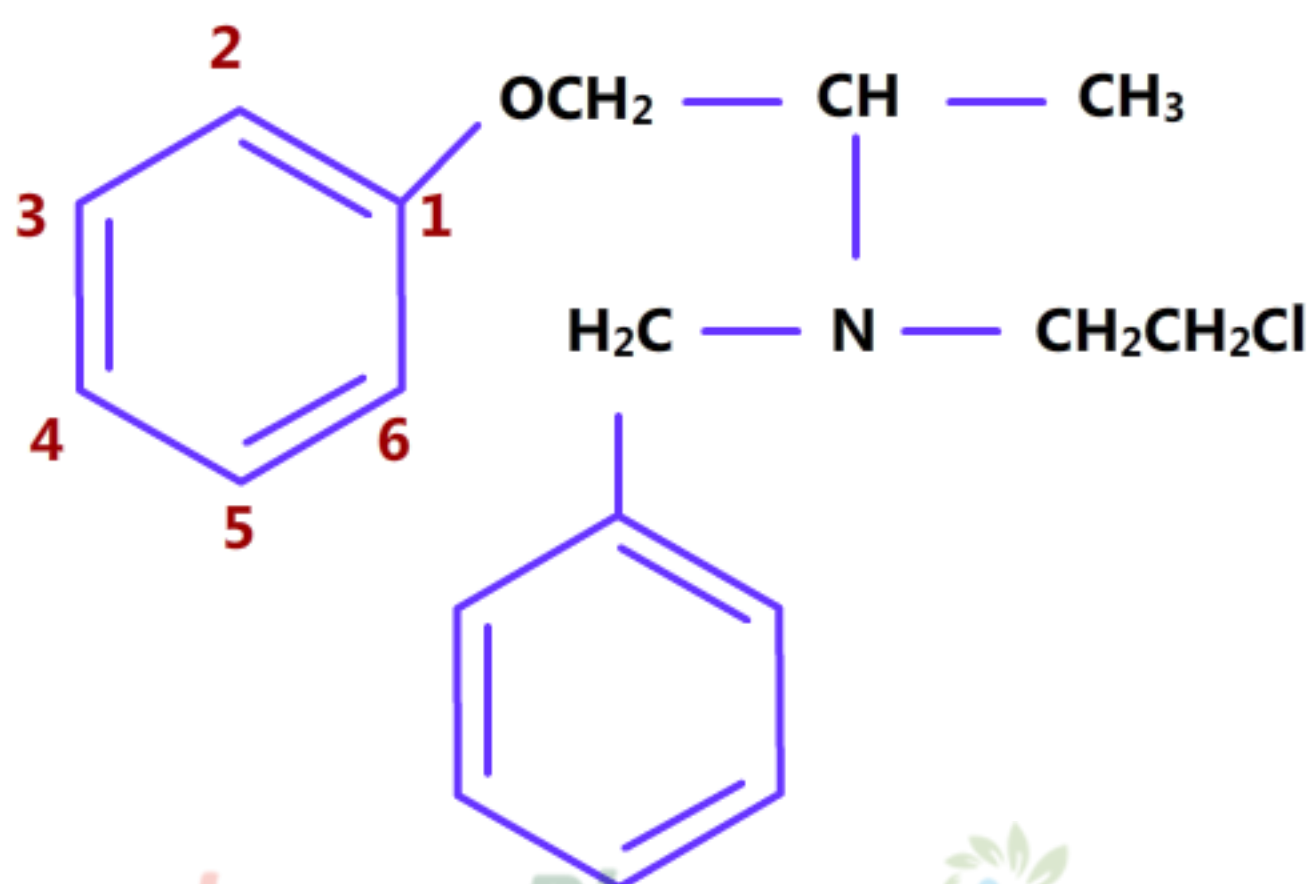


Fig 12: The structure of Phenoxybenzamine

- **IUPAC Name:** N-benzyl-N-(2-chloroethyl)-1-phenoxy-propane-2-amine
- **Molecular Formula:** C₁₈H₂₂ClNO

Mechanism of action :

- It widens blood vessels and relaxes muscles by blocking alpha receptors and due to widening of the blood vessels the blood pressure decreases.

Physical properties :

- Colour: White crystalline powder
- Odour: Aromatic odour
- Taste: Bitter in taste
- Melting point: 139 degrees
- Solubility: Soluble in water, methanol, chloroform, insoluble in ether

- Storage: store in airtight and light resistance closed containers at temp 15-20 degrees celsius
- Hygroscopic in nature

Physicochemical properties :

- Molecular weight: 303.8 g/mol
- Dihydroergotamine can be synthesised from reduction of the double bond of the ergoline ring of ergotamine at the 9-0 position.
- Melting point: 39 degrees
- Solubility: soluble in benzene

Uses :

- Used in the treatment of peripheral vascular disorders.
- It stimulates gastric secretion.
- It works by relaxing blood vessels.

Side effects :

- Tired feeling
- Upset stomach
- Blurred vision
- Dizziness
- Stuffy nose

Prazosin :

- It is derived from Quinazoline and it blocks alpha 1 receptor and it is a selective competitive blocker of alpha 1 receptor.

Structure :



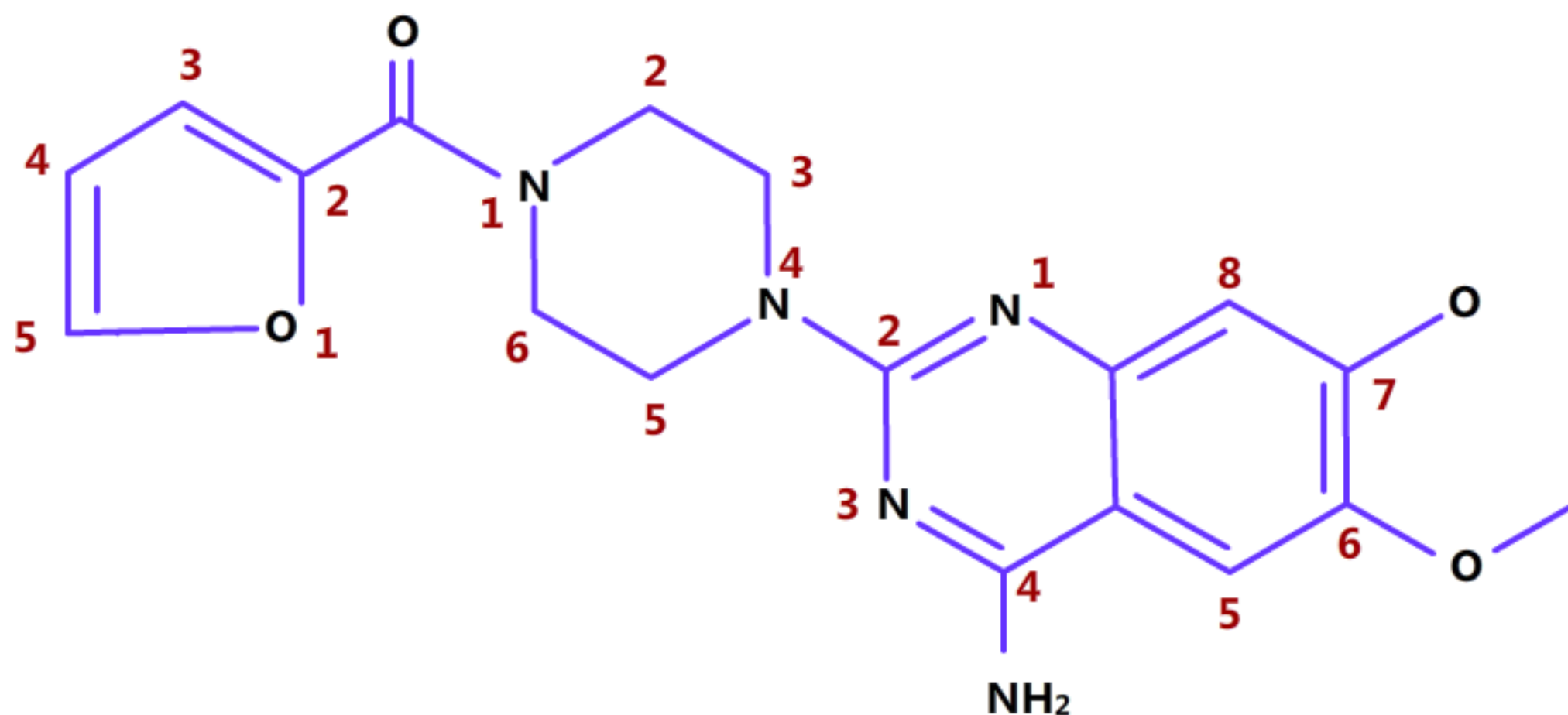


Fig 13: The structure of Prazosin

- **IUPAC Name:** 2-Furyl[4-(4-amino-6,7-dimethoxy-2-quinazolinyl)-1-piperazinyl]methanone
- **Molecular Formula:** C₁₉H₂₁N₅O₄

Mechanism of action :

- It selectively blocks alpha 1 receptors of the sympathetic system as a result neurotransmitter levels decrease and synthetic activities block and then reduce blood pressure.

Pharmacokinetic Properties :

- Prazosin was taken orally
- Widely distributed in the body
- It is metabolised in the liver
- It is well absorbed in the body
- This drug is eliminated by urine

Physical Properties :

- Colour: White or almost white crystalline powder
- Odour: Odourless
- Taste: Slight bitter
- Melting point: 96-97 degrees

- Solubility: slightly soluble in water and freely soluble in methanol ethanol, and insoluble in chloroform and acetone
- Storage: stored in an airtight and light resistance closed container

Uses :

- Used in the treatment of hypertension.
- Used in the treatment of Raynaud's disease.

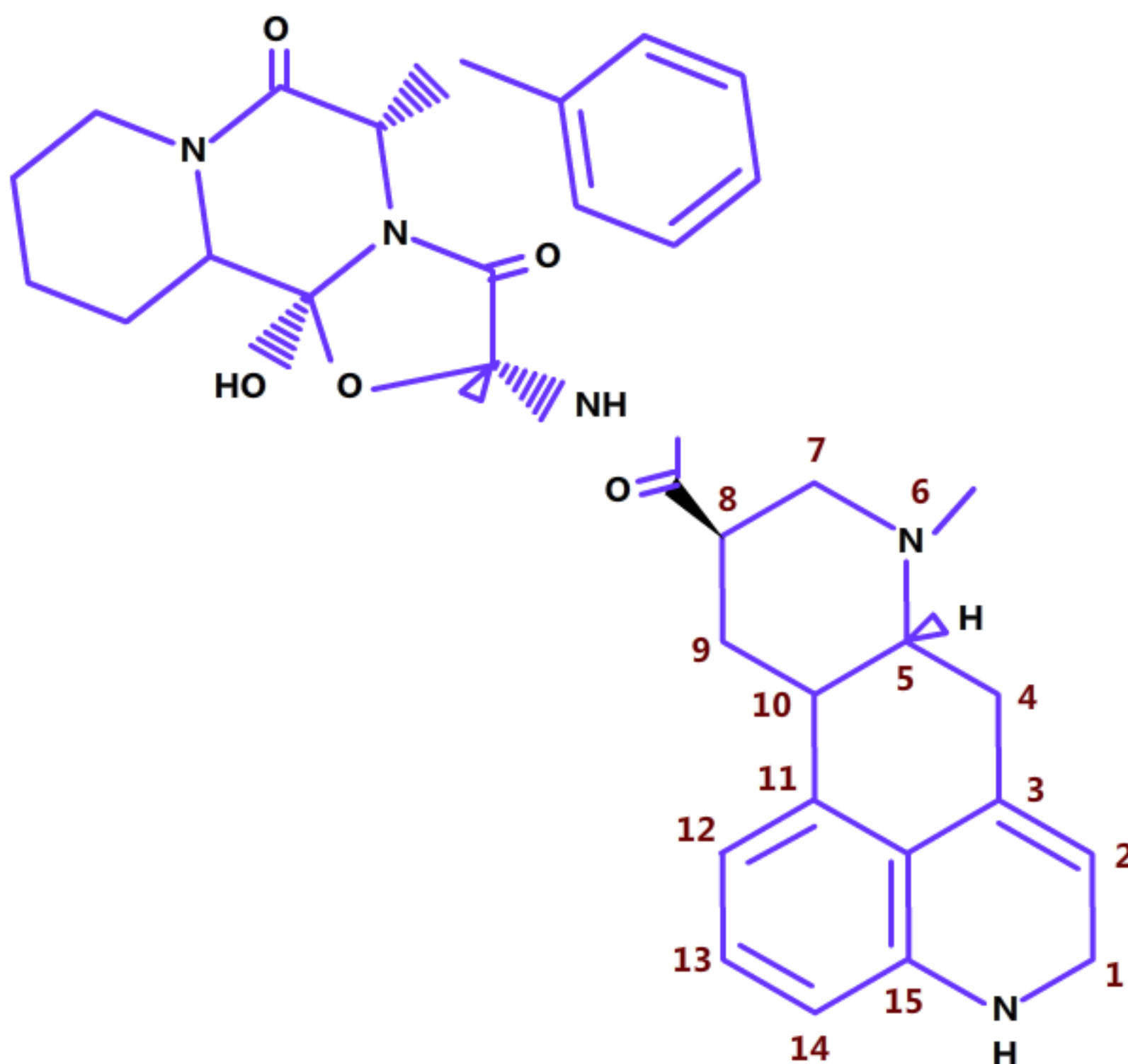
Side effects :

- Dizziness
- Headache
- Drowsiness
- Weakness
- Low energy
- Nausea

Dihydroergotamine :

Structure :





Dihydroergotamine

Fig 14: The structure of Dihydroergotamine

- **IUPAC Name:** [4-(4-Amino-6,7-dimethoxy-2-quinazolinyl)-1-piperazinyl](2-furyl)methanone
- **Molecular Formula:** $C_{33}H_{37}N_5O_5$

Mechanism of action :

- DHE has several proposed mechanisms which may contribute to its therapeutic efficacy as an abortive therapy in migraines.
- Firstly, DHE's agonist action on 5-hydroxytryptamine (5HT) $1b$ receptors in the smooth muscle of the cranial vasculature may provide relief via vasoconstriction of the blood vessels which typically become dilated due to the release of CGRP during migraine attacks.
- 3,4 DHE's off-target action at alpha-adrenergic receptors may further contribute via this mechanism.

Uses :

- Used in the treatment of Cluster headaches.
- Used in the treatment of migraines.

Properties :

- It acts as a serotonergic agonist, non-narcotic analgesic, a vasoconstrictor agent, dopamine agonist, sympatholytic agent.

Synthesis :

- Dihydroergotamine can be synthesised from reduction of the double bond of the ergoline ring of ergotamine at the 9-10 position.

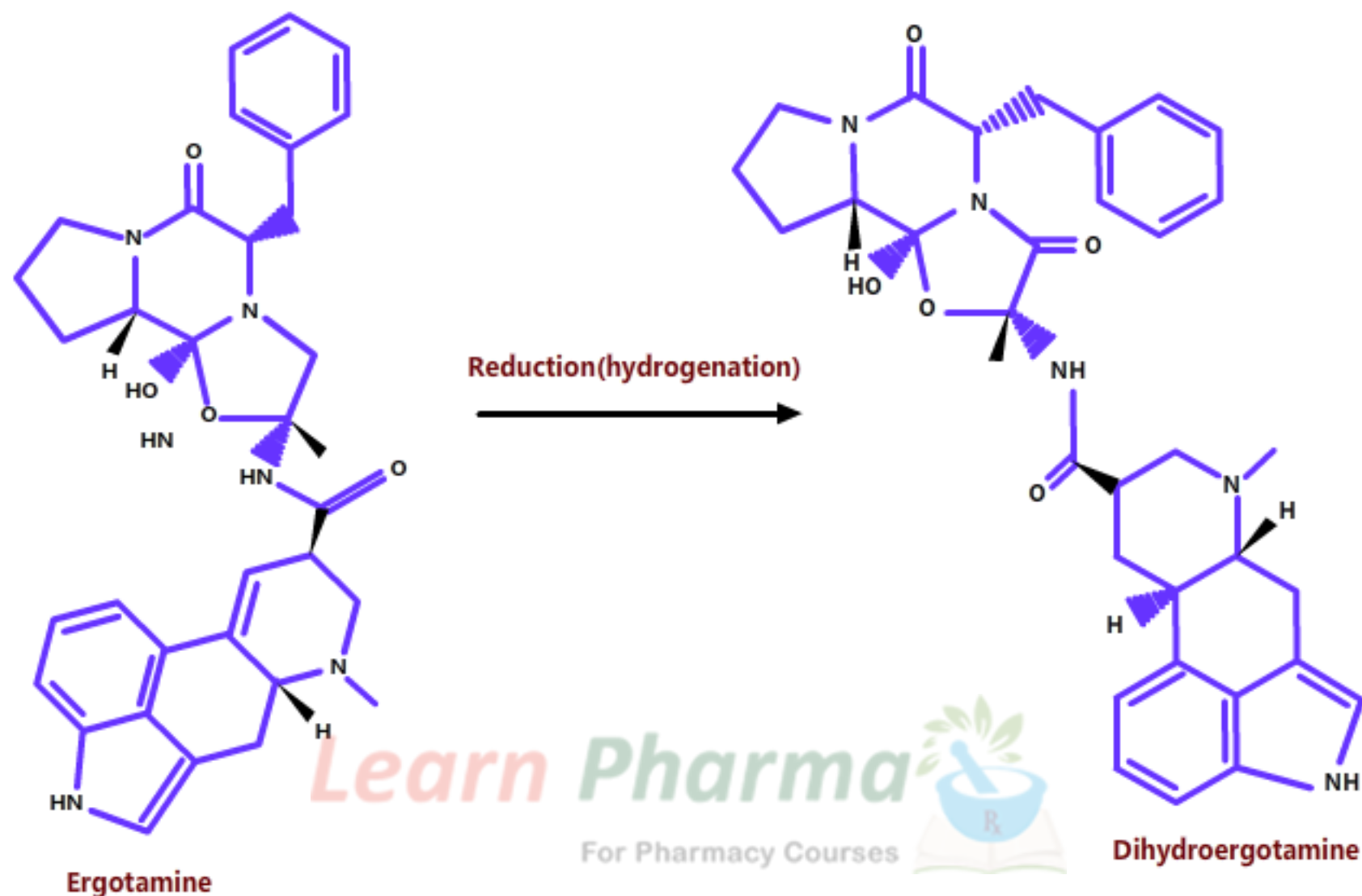


Fig 15: The synthesis of Dihydroergotamine

Side effects :

- Taste Disorder
- Dizziness
- Increased sweating
- Numbness
- Molecular weight : 583.7**

Side effects :

- Dizziness
- Drowsiness
- Vomiting

Diarrhoea

Methysergide :

- It is an ergot derivative related to lysergic acid diethylamide, its antagonist serotonin effect in blood vessels and gastrointestinal smooth vessels and also consists of a few properties of ergot alkaloids.

Structure :

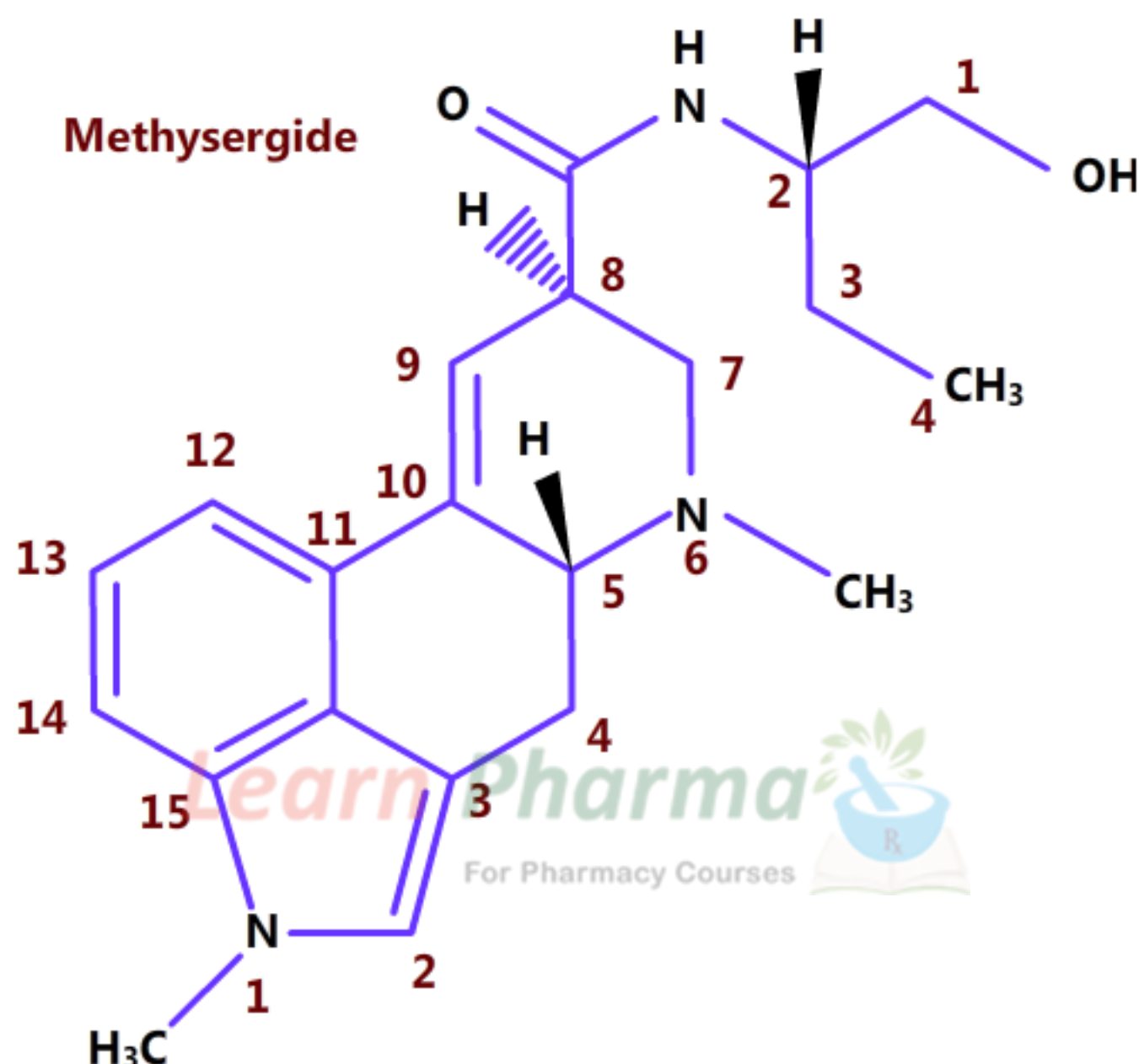


Fig 16: The structure of Methysergide

- IUPAC Name:** N-[(2S)-1-Hydroxy-2-butanyl]-1,6-dimethyl-9,10-didehydro ergoline-8-carboxamide
- Molecular Formula:** $C_{21}H_{27}N_3O_2$

Mechanism of action :

- It possesses some alpha adrenergic blocking activity and it is a serotonin antagonist acting on CNS and causing vasoconstriction by directly stimulating the smooth muscles.

Side effects :

- Nausea
- Vomiting

- ☐ Heartburn
- ☐ Drowsiness
- ☐ Dizziness

Uses :

- ☐ It is used in the treatment of vascular headaches.

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Beta Adrenergic blockers

Introduction :

- Beta Adrenergic blockers are the drugs which are used in the medication to reduce blood pressure by blocking the effects of the neurotransmitter adrenaline or noradrenaline.
- The classification of Beta adrenergic antagonist drugs are characterised into selective and non selective beta blockers.
 - Non- selective(beta 1 and beta 2) antagonists: propranolol, sotalol, Timolol, Pindolol, Labetalol, Carvedilol.
 - Selective (Beta 1) antagonists: metoprolol, Atenolol, Acebutolol, Bisoprolol, Esmolol, betaxolol, celiprolol, Nebivolol.

Mechanism of action :

- The heart pumps blood around your body through a network of blood vessels known as the circulatory system.
- Blood supplies the cells and tissues with the oxygen and nutrients.
- Beta blockers like propranolol, atenolol etc are administered that block the actions of the neurotransmitters called adrenaline and noradrenaline.
- These neurotransmitters increase the heart rate and force of contraction, so blocking beta receptors decreases the heart rate which reduces blood pressure and the workload on the heart.

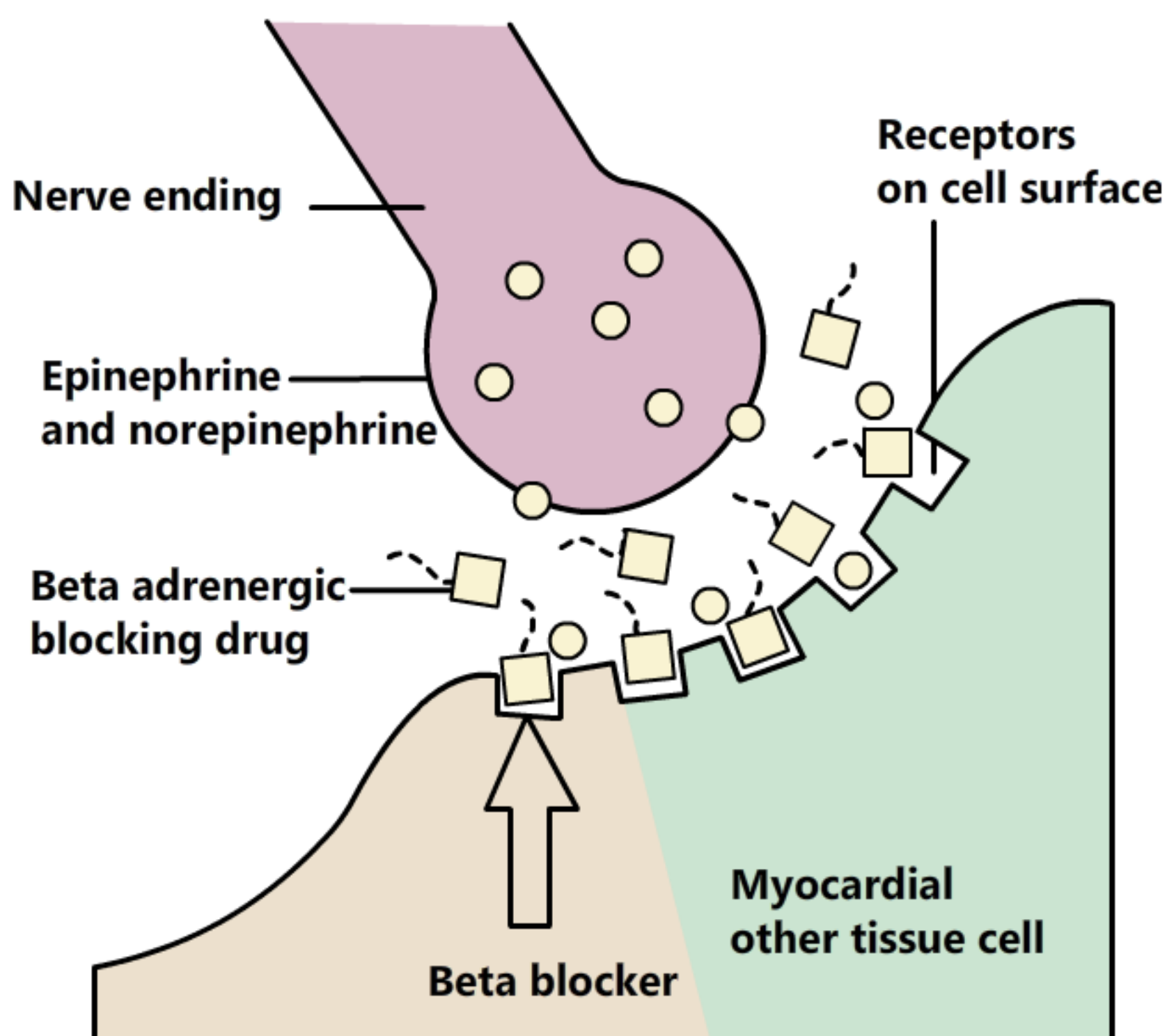


Fig 17: Mechanism of Beta blocker

- Beta receptors are of three types: beta 1, beta 2, beta 3 receptors.
- Beta 1 receptors are mainly found in the heart and also in the kidneys.
- Beta 2 receptors are found in the lungs, GI system, vascular muscles, skeletal muscle and ciliary body of the eye.
- Beta 3 receptors are found in fatty \ Adipose tissue in the body.

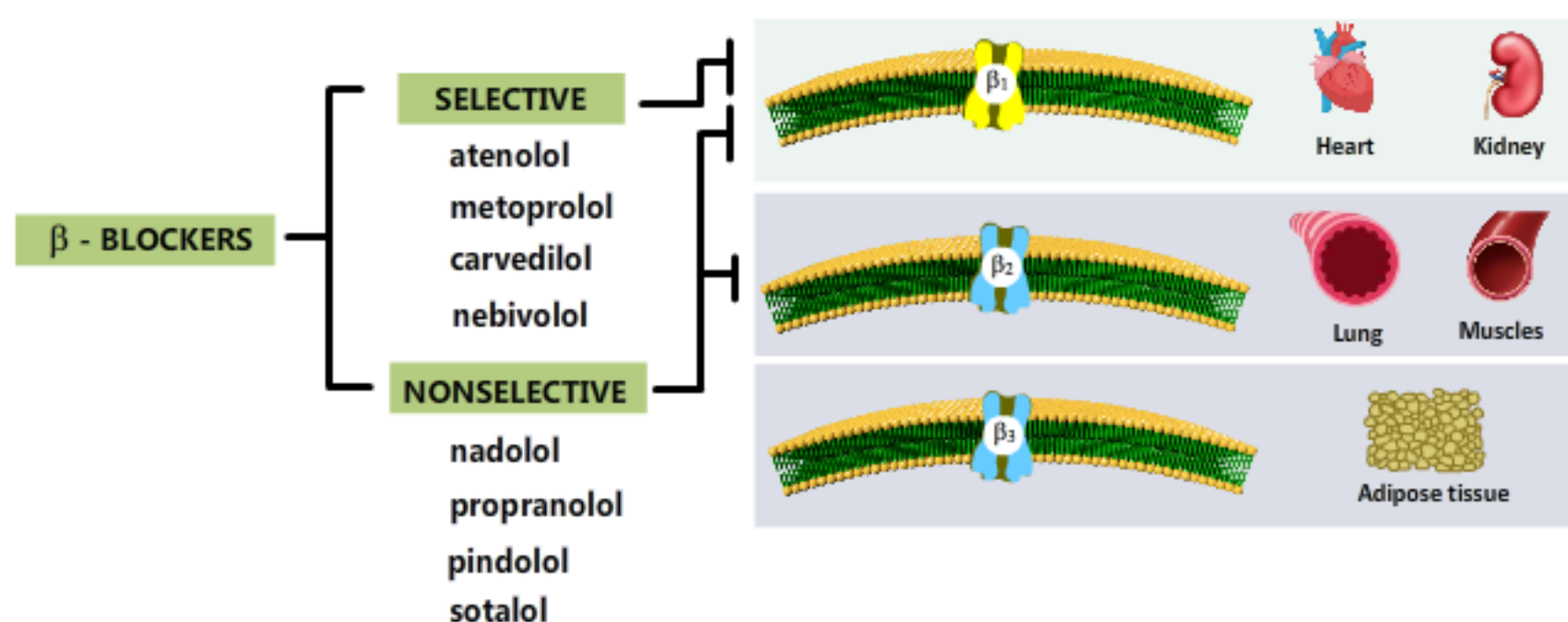


Fig 18: The types of beta blockers

Classification of beta blockers :

- Selective: It targets only specific receptors either beta 1 or beta 2 or beta 3 receptors.

- Non selective: it targets both beta 1 and beta 2 receptors.
- These selective and non selective drugs end with the lol suffix.
- Selective beta receptors:** Atenolol, Esmolol, Metoprolol, Bisoprolol, Acebutolol.
- Non Selective beta receptors:** Sotalol, Timolol, Propranolol, Nadolol, carteolol, penbutolol.

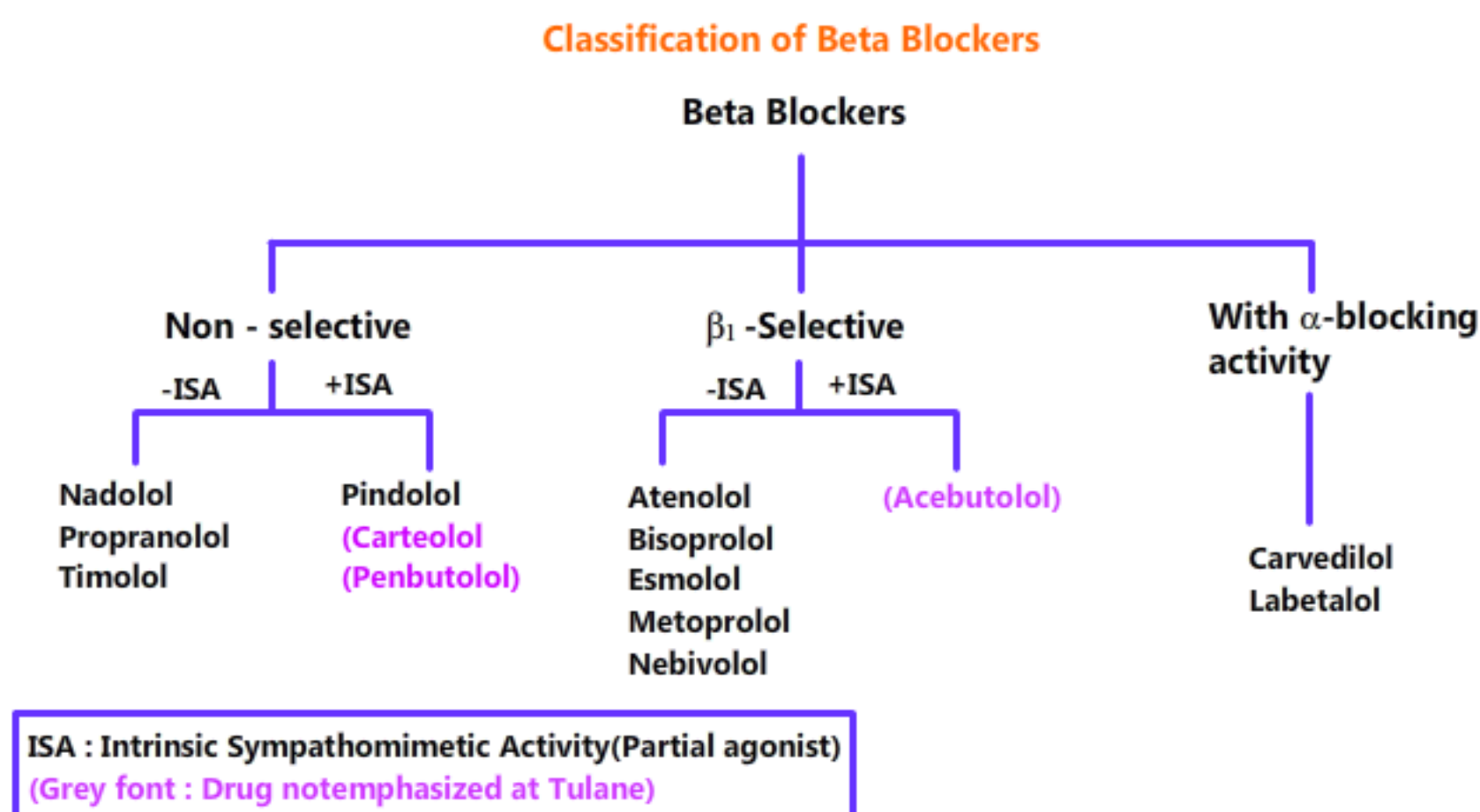


Fig 19: Classification of Beta blockers

SAR of Beta Adrenergic blockers :

- The general structure of beta blocker for Propranolol is:

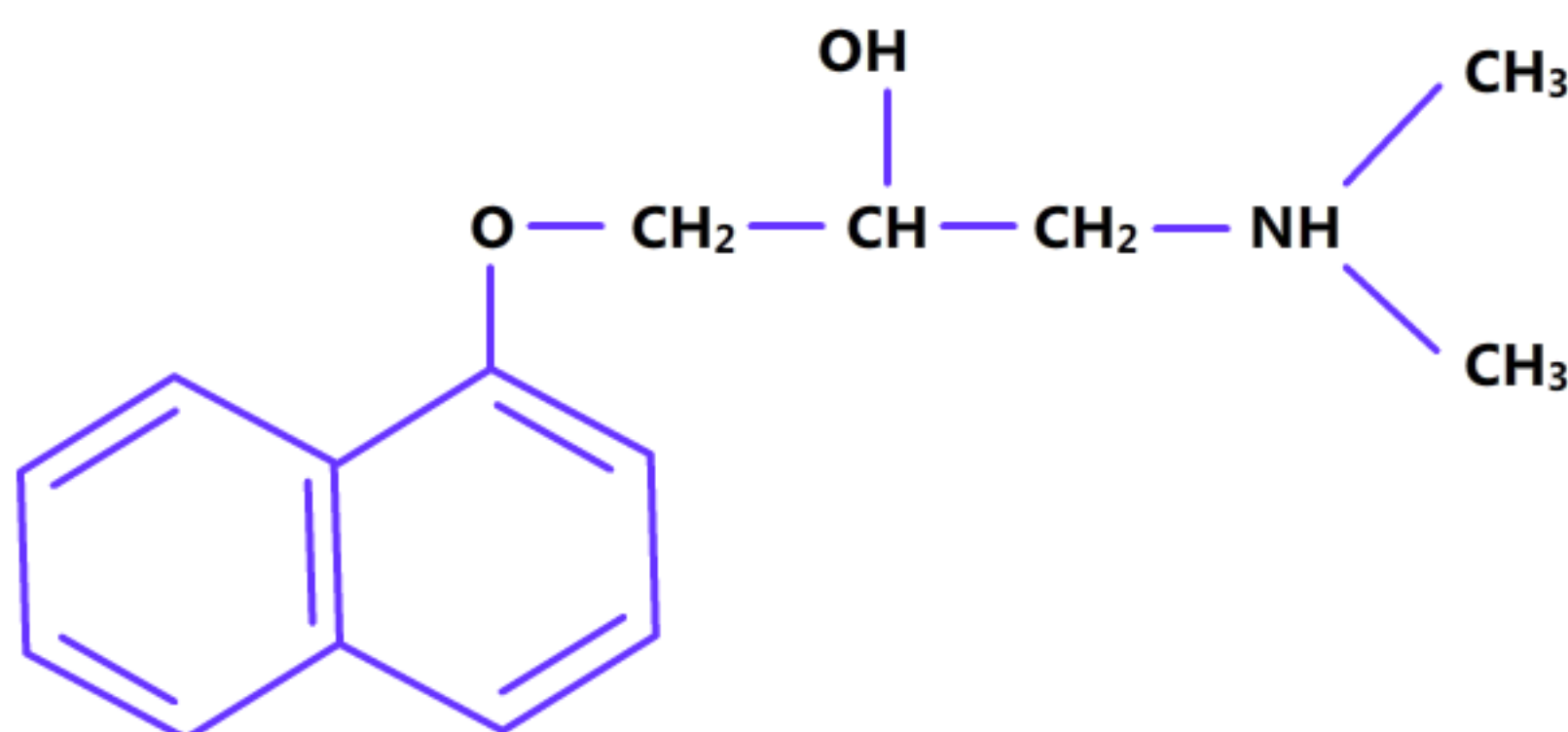


Fig 20: Structure of Propranolol

- In the basic structure propranolol the structure consists of Aromatic ring, carbonyl group and amino group.
- When the alkenyl and alkanol oxy group is present in the ortho position on phenyl ring shows beta antagonist activity.

- Example: Alprenolol, oxprenolol.

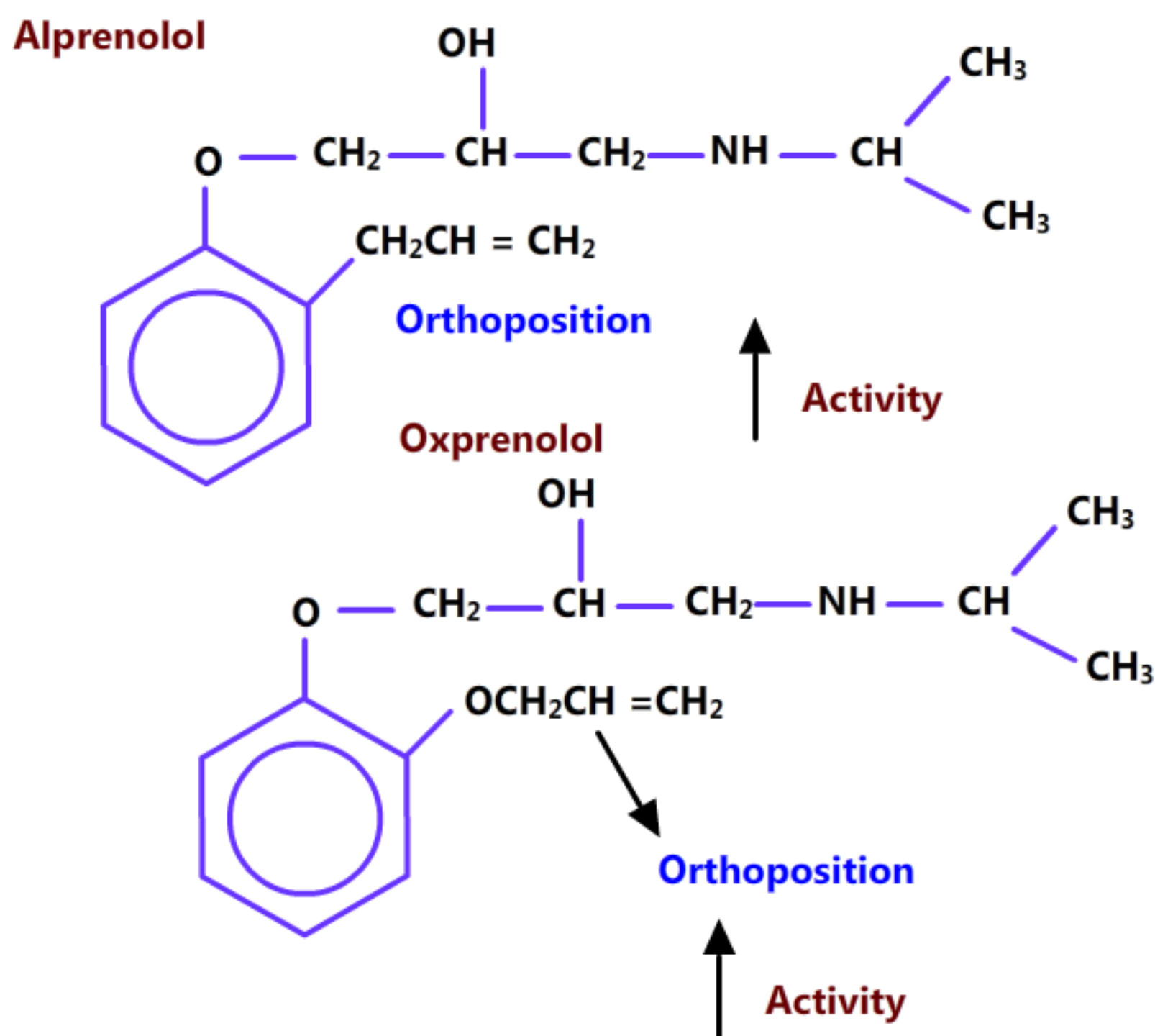
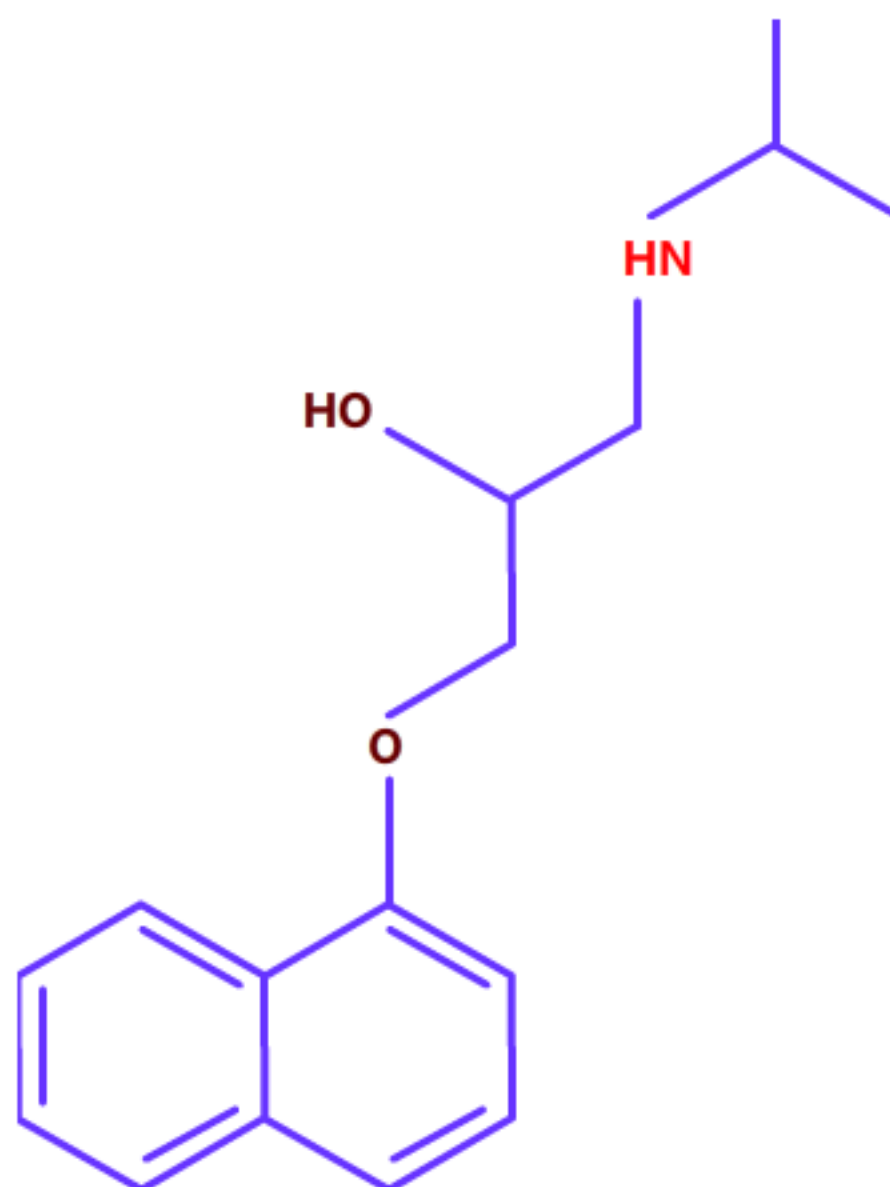


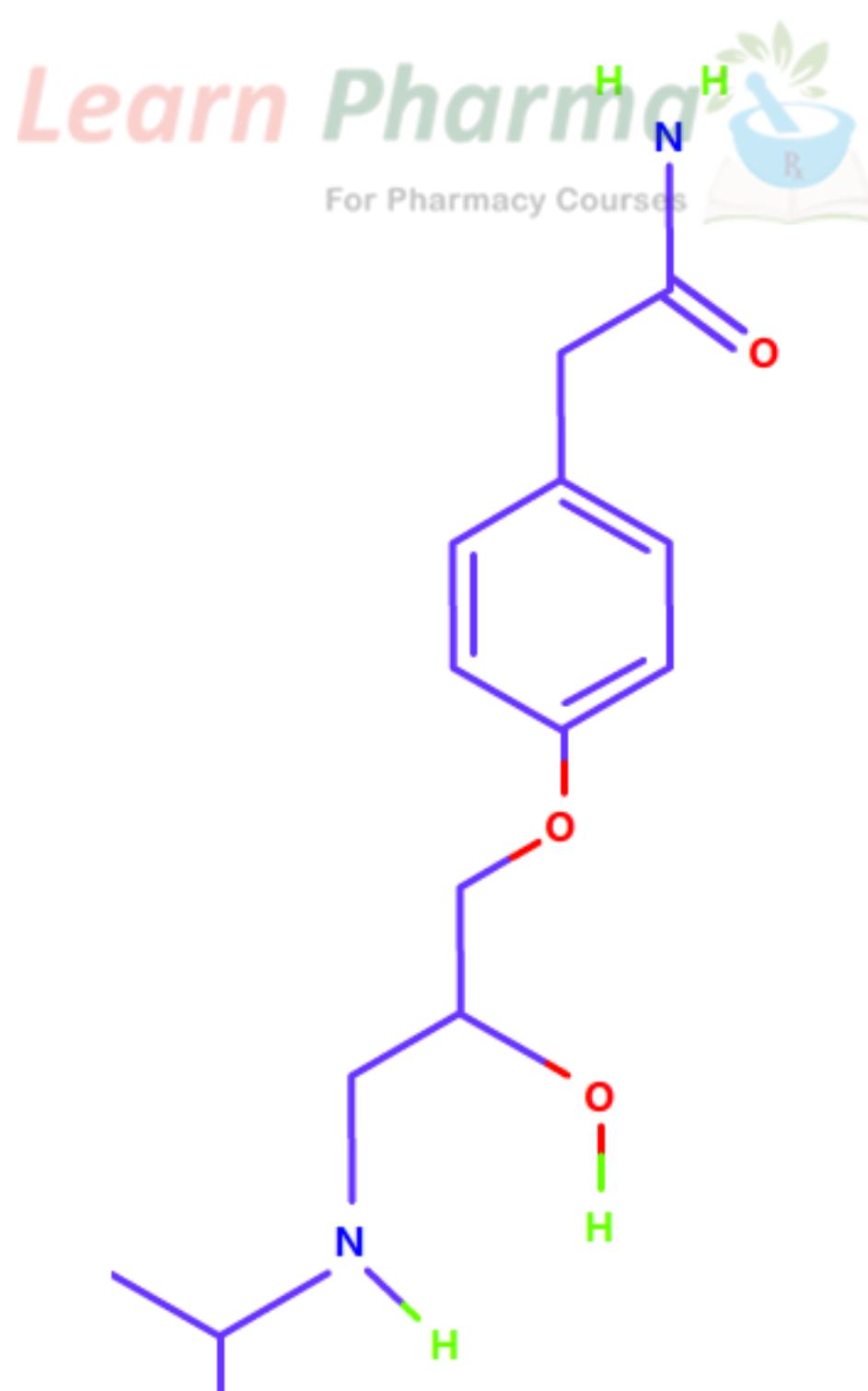
Fig 21: The structure of Alprenolol and oxprenolol

- By addition of the -OH group in phenyl ring which does not show antagonist activity.
- The phenyl ring is substituted with the naphthyl ring; it is a non-selective beta blockers.
- Example: propranolol.



- The para substitution group in the aromatic ring shows more activity than compared to the meta position.

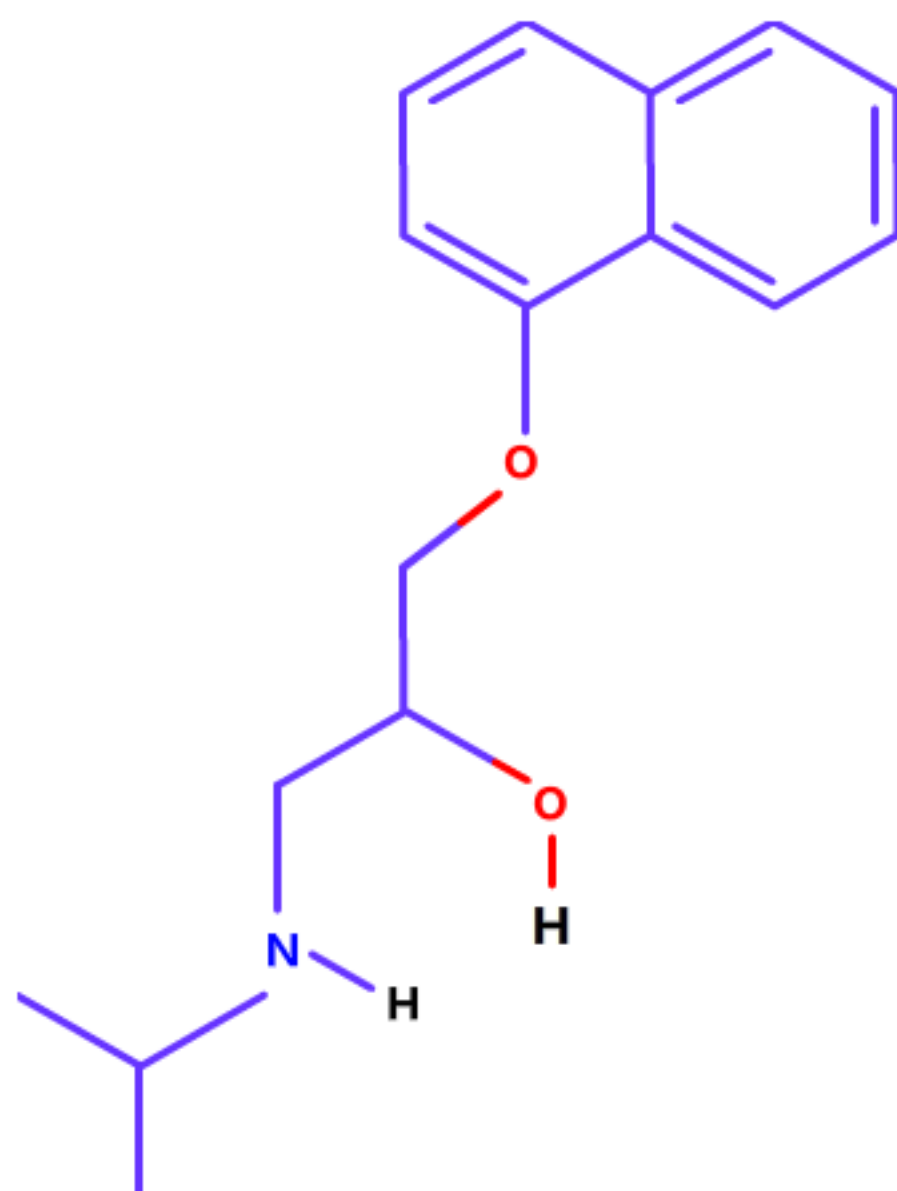
Example: Atenolol.



Substitution of Carbon chain :

- The $-OCH_2$ group is placed between the aromatic ring and ethanolamine side chain; it increases the activity.

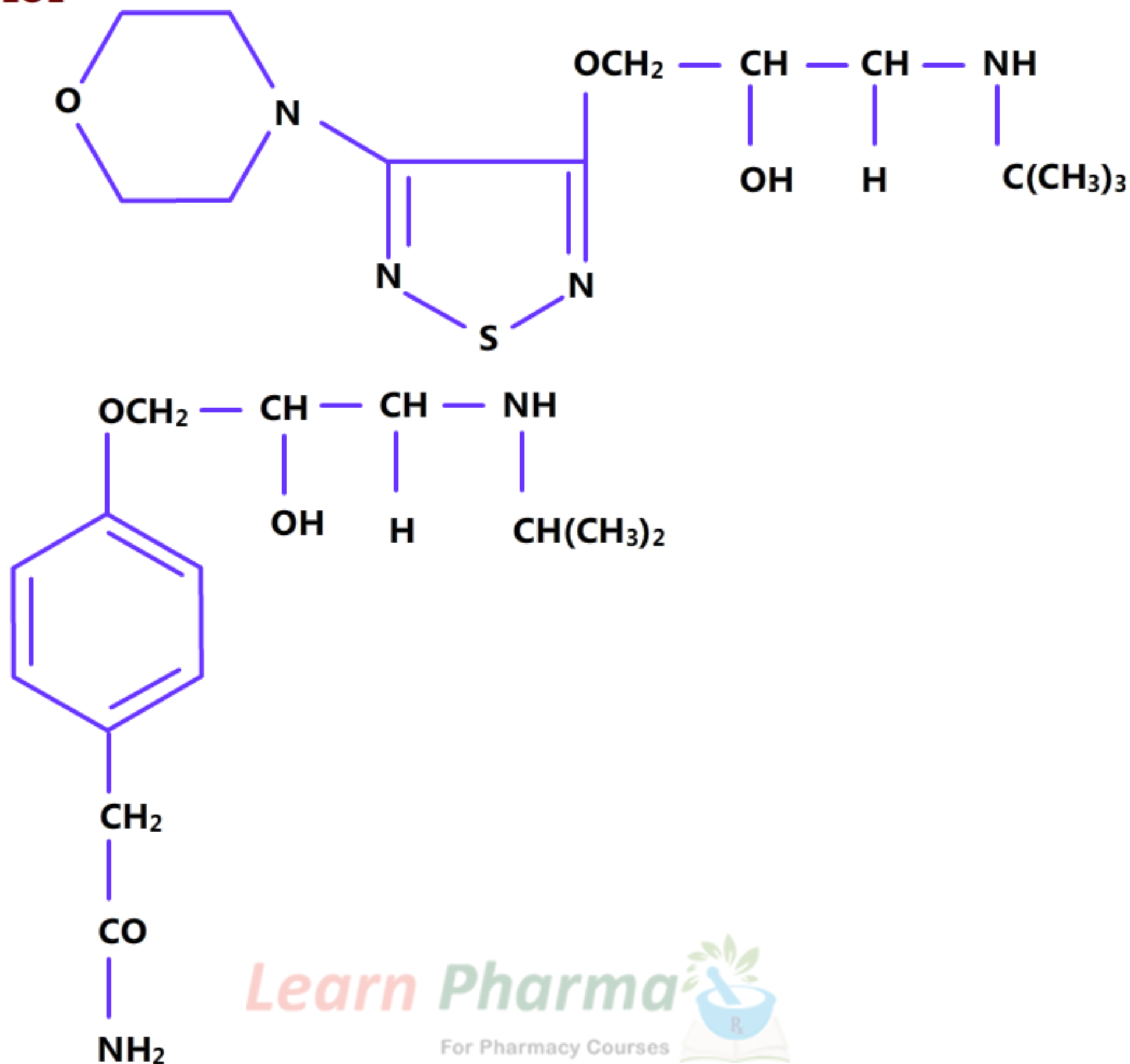
- The $-OCH_2$ is replaced by H then the compound is a non selective beta blocker.
- Examples: propranolol.



Substitution of Amino group :

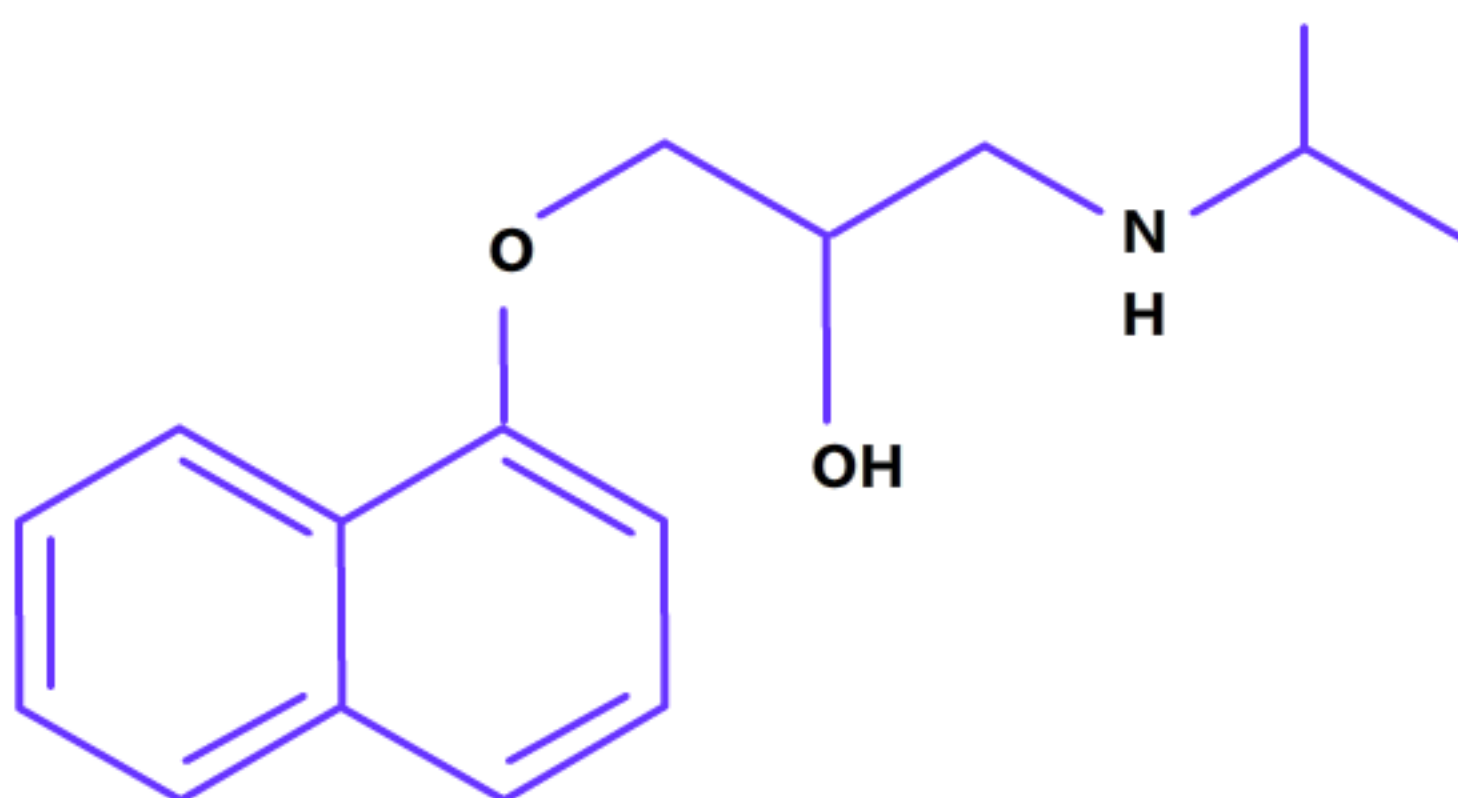
- The substitution of the amino group shows nucleophilicity and increases the activity.
- Example: Atenolol, Timolol.

TIMOLOL



Propranolol :

Structure :



Propranolol

Fig 22: The structure of Propranolol

IUPAC Name : 1-(Isopropylamino)-3-(1-naphthyloxy)-2-propanol

Properties :

- It is a stable, white, crystalline substance that is easily soluble in both water and ethanol.
- It has a molecular weight of 295.80.

Synthesis of Propranolol :



- Propranolol as an example of a non-selective antagonist at Beta-adrenergic receptor and is used in treatment of hypertension, cardiac arrhythmias and congestive heart failure.
- Now let us see the synthesis of Propranolol step by step.
- The synthesis of Propranolol begins with the starting material, 1-naphthol is treated with aqueous potassium hydroxide forming a potassium salt of naphthalene-1-olate and subsequent loses of water.
- The second step involves potassium salt of naphthalene-1-olate treated with epichlorohydrin.
- During reaction, the naphthoxy anion attacks on the carbon atom of epichlorohydrin forming an intermediate 2((naphthalene-1-yl-oxy)methyl) oxirane.
- Meanwhile the potassium ion from potassium salt of naphthalene-1-olate and the chlorine from epichlorohydrin together are eliminated as potassium chloride.
- In the final step, an intermediate further added with isopropyl amine undergoes a nucleophilic addition reaction.
- During this reaction isopropyl amine opens the oxirane ring by attacking on the electrophilic carbon atom of 2((naphthalene-1-yl-oxy)methyl) to form the target molecule, propranolol.

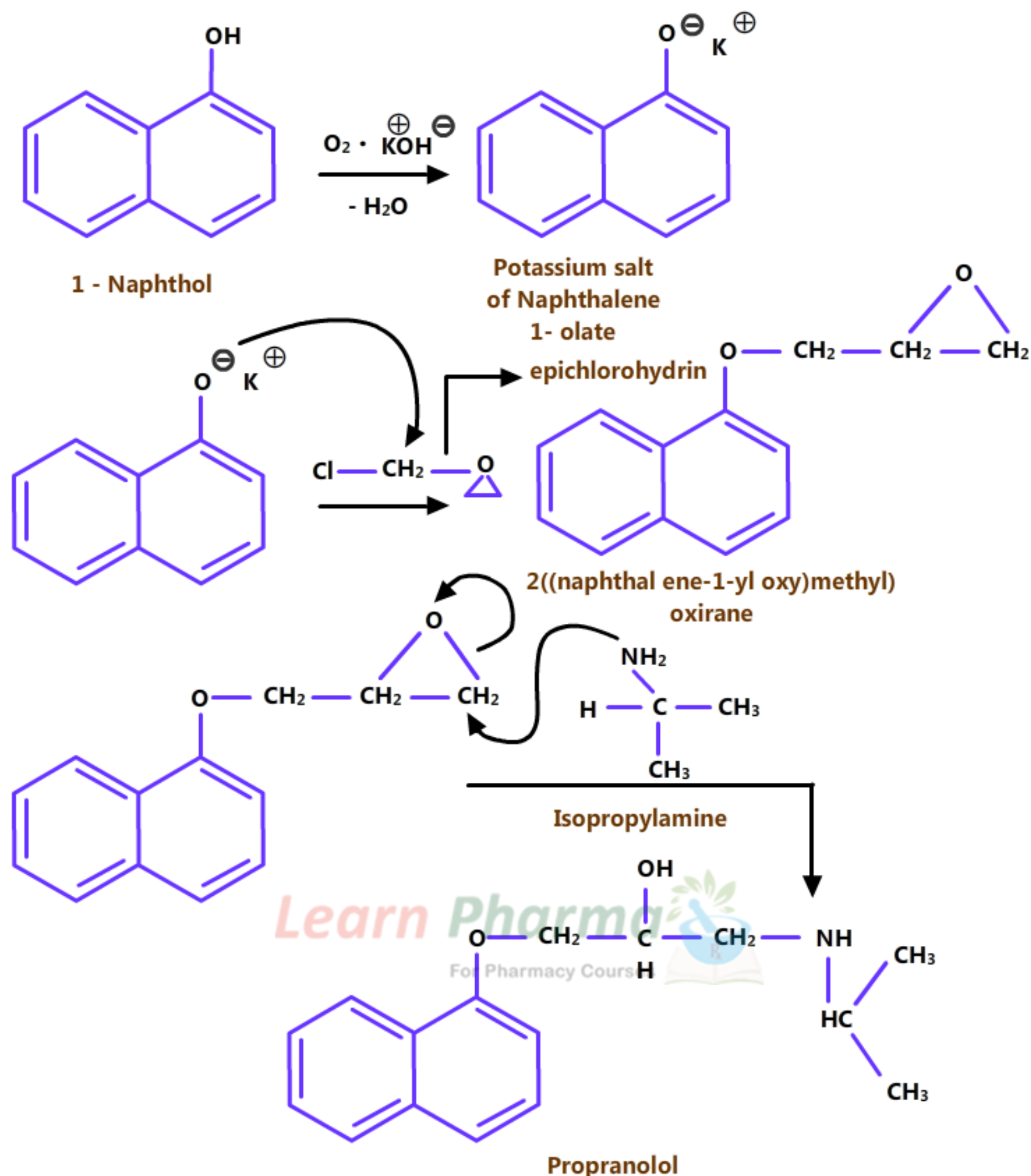


Fig 23: Synthesis of Propranolol

Mechanism :



- It is a nonselective β -adrenergic receptor antagonist.
- Blocking of these receptors leads to vasoconstriction, inhibition of angiogenic factors like vascular endothelial growth factor and basic growth factor of fibroblasts, induction of apoptosis of endothelial cells, as well as down regulation of the renin-angiotensin-aldosterone system.

SAR of Propanol :

- If the side chain of the structure of propanol increases then the binding does not take place towards the receptor site.

- The group of aryloxy propylamines has more potency compared to side chain permits and has a close overlap with the aryl ethionamide side chain.

Metabolism :

- It undergoes side chain oxidation to α -naphthoxylactic acid, ring oxidation to 4'-hydroxypropranolol or glucuronidation to propranolol glucuronide.
- It can also be N-desisopropylated to become N-desisopropyl propranolol.
- 17% of a dose undergoes glucuronidation and 42% undergoes ring oxidation.

Uses :

- It is used to treat hypertension, angina, migraine, essential tremor, hypertrophic subaortic stenosis and pheochromocytoma, atrial fibrillation, myocardial infarction

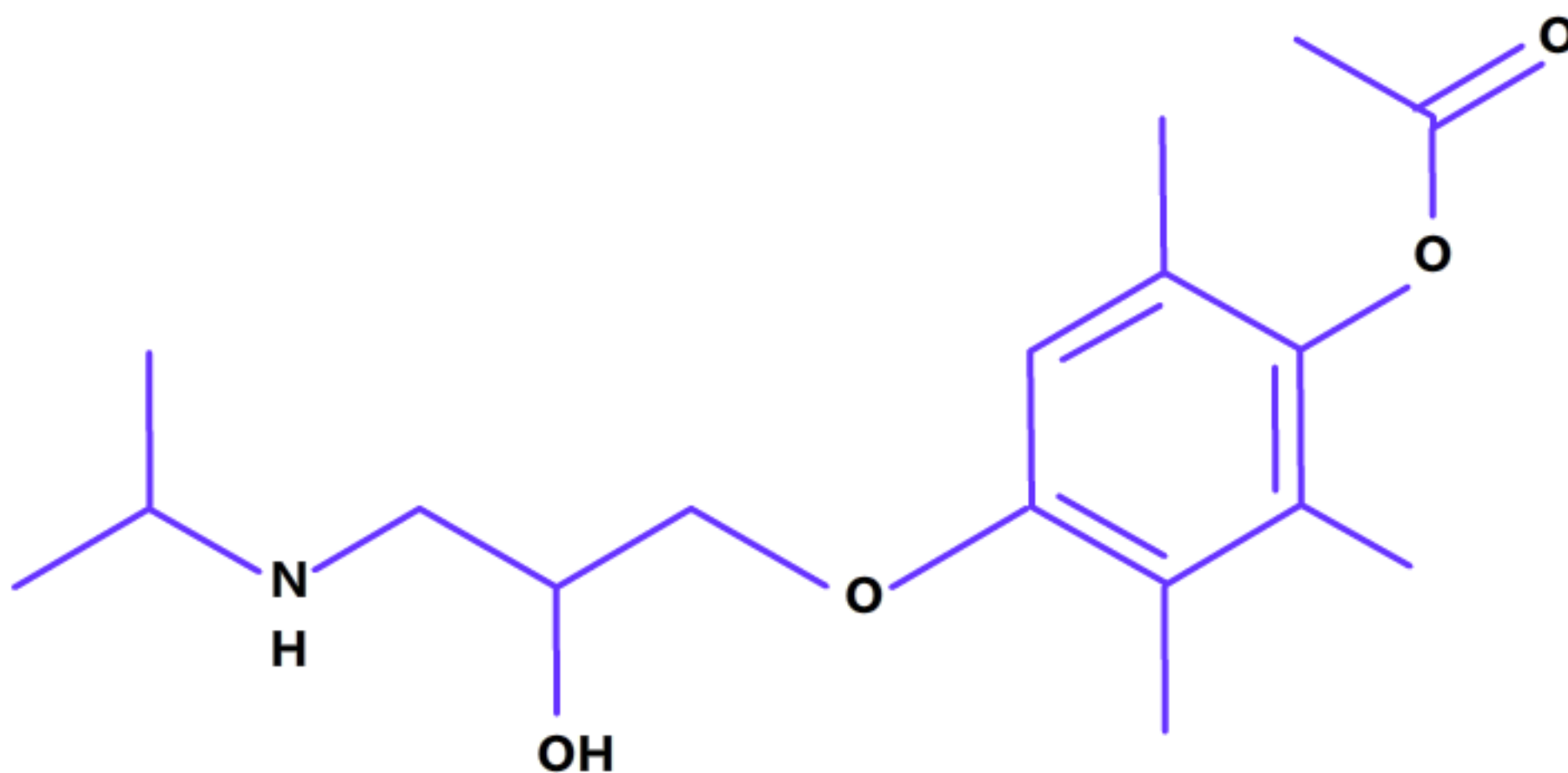
Adverse effects :

- Feeling or being sick (nausea or vomiting) or diarrhoea
- Stomach pain
- Headaches
- Feeling tired, dizzy or weak
- Cold fingers or toes

Metipranolol :

Structure :





Metipranolol

Fig 24: Structure of Metipranolol

- **IUPAC Name:** 3-(Propan-2-ylamino)propane-1,2-diol

Properties :

- Physical appearance : Present in solid form
- Melting point : 106°C
- Molecular weight : 309.4 g/mol
- Solubility : 589 mg



Mechanism :

- Metipranolol binds the beta1 and beta2 adrenergic receptors, the mechanism of metipranolol's action is not known. It has no significant intrinsic sympathomimetic activity and has only weak local anaesthetic (membrane-stabilising) and myocardial depressant activity.

Uses :

- It is used for the reduction of elevated intraocular pressure in patients with ocular hypertension or open-angle glaucoma.

Adverse effects :

- Headache
- Confusion

- Slow heart rate
- Shortness of breath
- Swelling and discomfort around the eyes
- Bronchospasm
- Sudden numbness and weaknessWeight gain
- Watering of eyes
- Blurred vision
- Insomnia

Atenolol :

Structure :

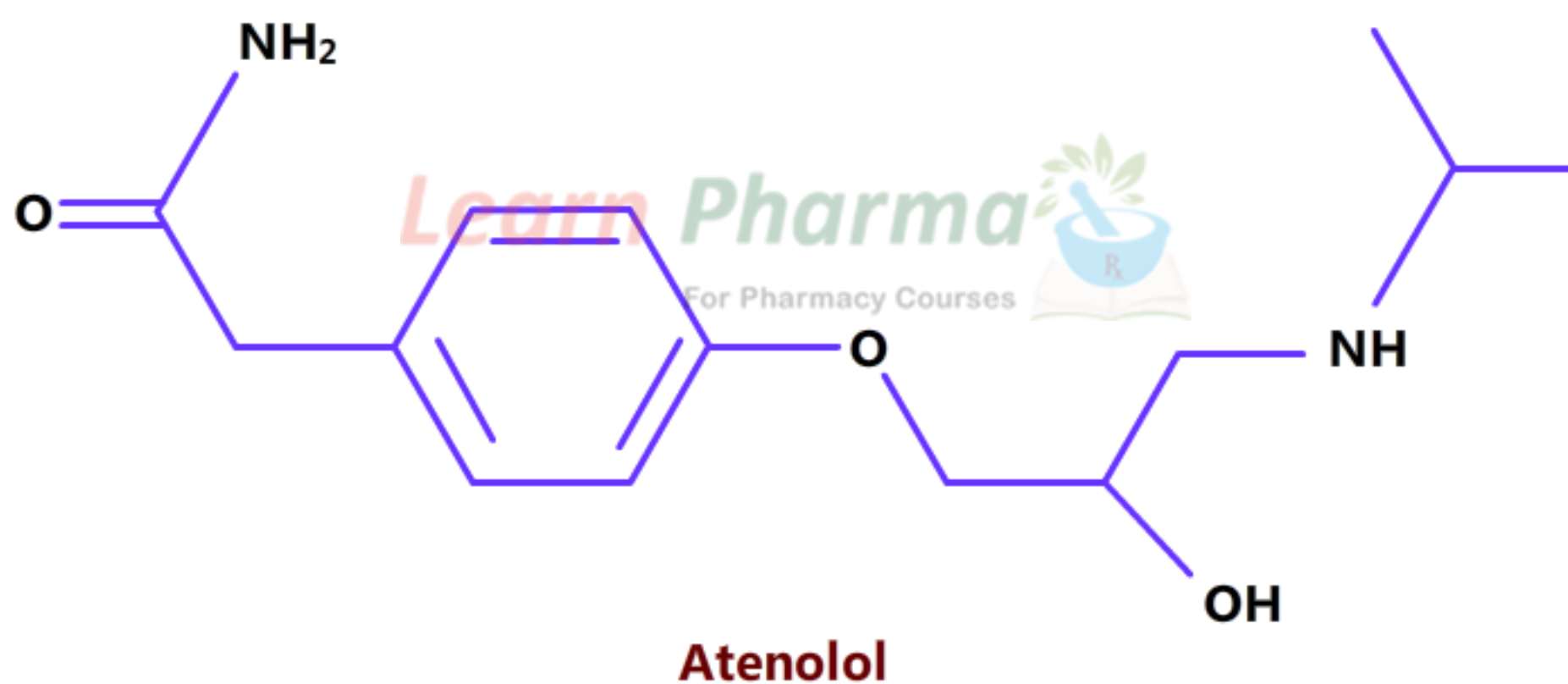


Fig 25: Structure of Atenolol

- **IUPAC Name :** (RS)-2-{4[2Hydroxy3(propan2ylamino)propoxy]phenyl}acetamide

Properties :

- This is an beta selective adrenoceptor blocking drug of membrane stabilising activity.

Mechanism :

- Atenolol acts by selectively binding to the beta-1 adrenergic receptors found in vascular smooth muscle and the heart, blocking the positive inotropic and chronotropic actions of endogenous catecholamines such as isoproterenol, norepinephrine and epinephrine, thereby inhibiting sympathetic stimulation.

- This action lowers myocardial contractility, lowers heart rate and lowers blood pressure. Atenolol, on the other hand can raise end-diastolic pressure and left ventricular fibre lengths in those with heart failure, which in turn raises oxygen demand.

Metabolism :

- It is metabolism in the liver.
- The sole non-conjugated metabolite is the product of a hydroxylation reaction at the carbon between the amide and benzene groups.
- The only other metabolite to be confirmed is a glucuronide conjugate. These metabolites make up 5-8% and 2% of the renally excreted dose with 87-90% appearing as unchanged drug. The hydroxylated metabolite exerts 1/10th the beta-blocking activity of atenolol.

Uses :

- It is used in the management of hypertension and chronic angina.

Adverse effects :

- Bradycardia, diarrhoea, dizziness, constipation, confusion, dyspnea, headache, heart failure, erectile dysfunction, nausea, fatigue, paraesthesia, peripheral coldness, rash, sleep disorders, syncope, visual impairment and bronchospasm.

Betaxolol :

Structure :

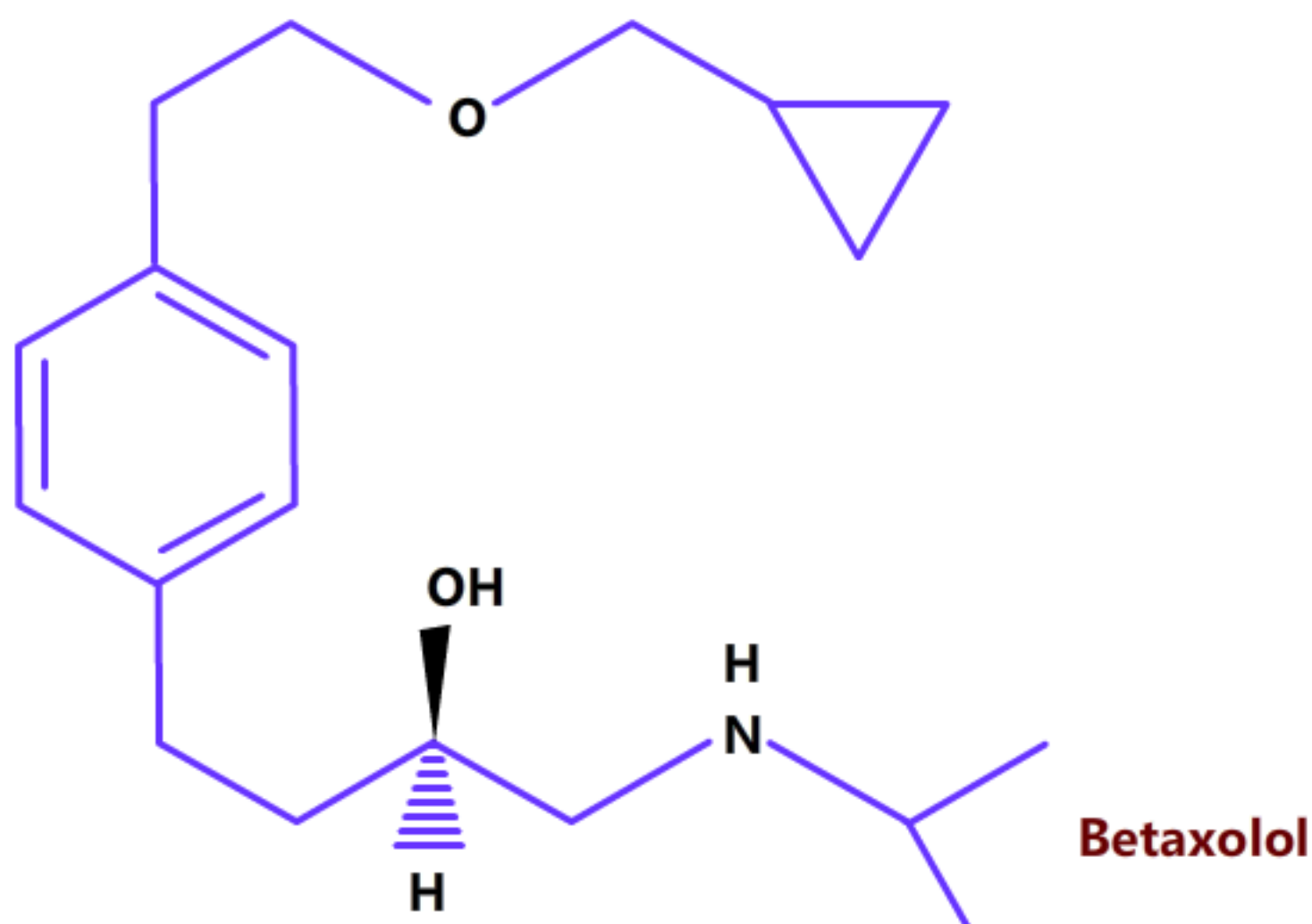


Fig 26: Structure of betaxolol

- **IUPACName :**
(RS)1{4[2(cyclopropylmethoxy)ethyl]phenoxy}3(isopropylamino)propan-2-ol

Properties :

- Physical appearance : Solid
- Molecular weight : 307.4 g/mol
- Solubility : 451 mg
- Melting point : 71°

Mechanism :

- Betaxolol specifically binds to and blocks the β_2 -receptors found in the smooth muscles of the heart and arteries.
- As a result, there is a decrease in heart rate, cardiac output, systolic and diastolic pressure, as well as reflex orthostatic hypotension.
- By competitively inhibiting β_2 -adrenergic receptors in the smooth muscles of the bronchial and vascular systems, betaxolol also produces bronchospasm.

Metabolism :

- Around 15% of the dose taken is excreted as an unmodified medication, with the remaining metabolites having little to no impact on the clinical outcome.

Uses :

- It is used for treatment of high blood pressure, glaucoma, prevention of heart attacks and prevention of kidney problems.

Adverse effects :

- Diarrhoea
- Nausea
- Stomach upset
- Decreased libido
- Insomnia
- Headache

- Drowsiness
- Lightheadedness
- Dizziness

Bisoprolol :

Structure :

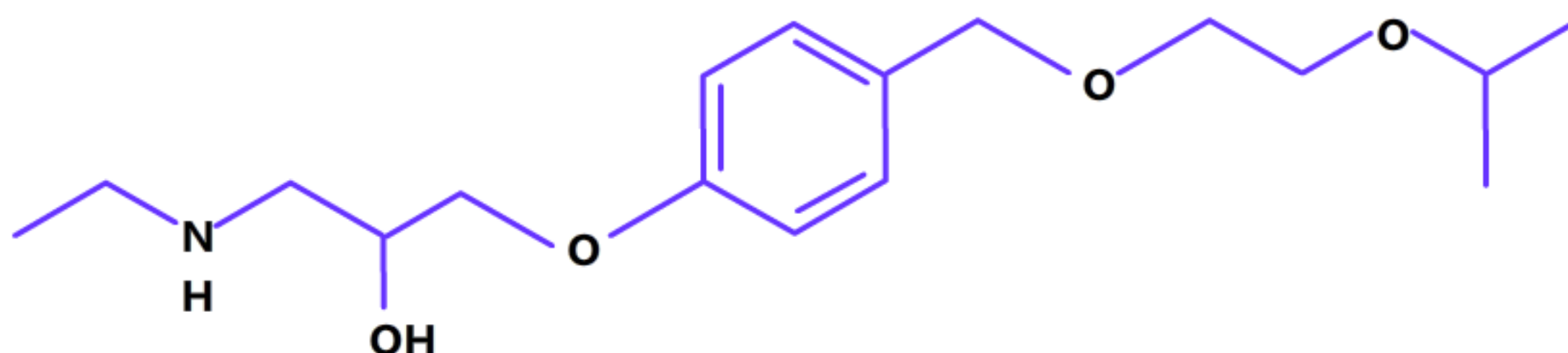


Fig 27: Structure of Bisoprolol

- **IUPAC Name :** (RS)1{4[(2Isopropoxyethoxy)methyl]phenoxy}3(isopropylamino)propan-2-ol



Properties :

- Physical appearance : Solid
- Molecular weight : 325.4 g/mol
- Melting point : 100°C
- Solubility : In water

Mechanism :

- It acts as an antagonist at β_1 -adrenergic receptors which results in the lowering of the blood pressure through reduction in the cardiac output. Moreover, it reduces the kidneys' production of rennin.

Uses :

- It is a cardioselective beta-blocker used in the treatment of hypertension.

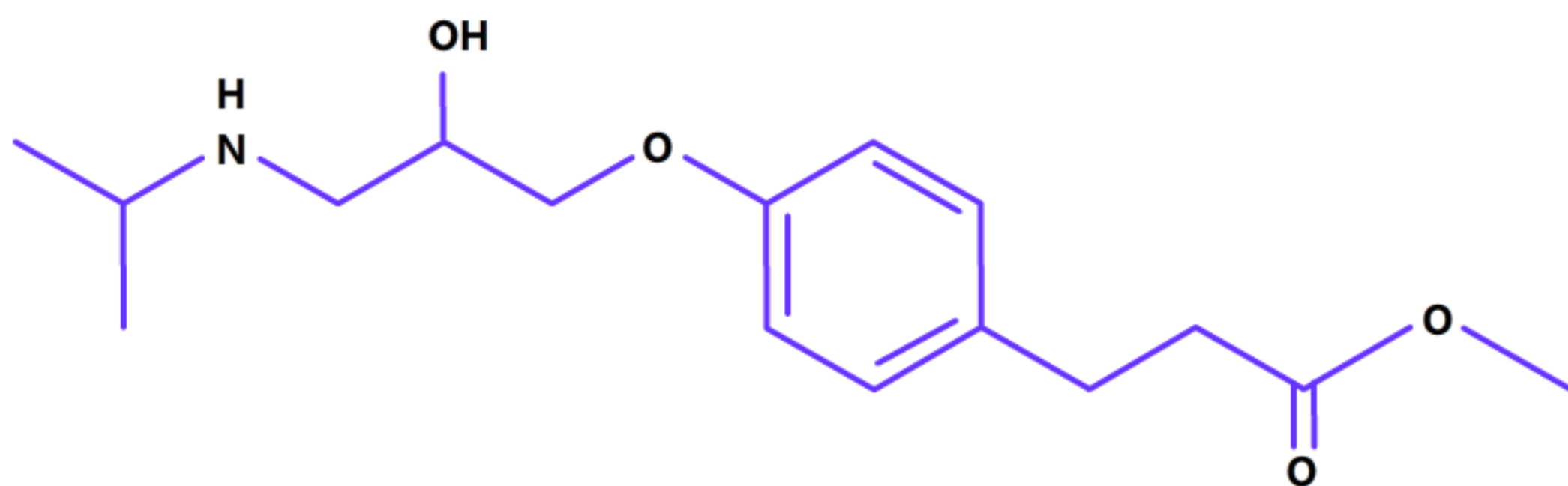
Adverse effects :

- Headache

- Cold or flu symptoms
- Swellings
- Weakness
- Diarrhoea
- Slower heart rate
- Allergic reactions like skin rashes
- Breathing problems
- Confusion
- Sweating
- Tremors
- Vomiting

Esmolol :

Structure :



Esmolol

Fig 28: Structure of Esmolol

- **IUPAC Name :** Methyl(RS)-3-{4-[2-hydroxy-3-(propan2ylamino)propoxy]phenyl} propanoate

Properties :

- Physical appearance : Solid
- Melting point : 90°C

- Solubility : Very soluble in hydrochloride salt
- Molecular weight : 295.37 g/mol

Mechanism :

- Esmolol suppresses the agonist effect of sympathetic neurotransmitters by competitively binding with β_1 -receptors in heart tissues.
- It inhibits adrenergic stimulation of the potential of the cardiac pacemaker and induces antiarrhythmic activity.

Metabolism :

- Esmolol is rapidly metabolised by hydrolyzing the ester bond, mostly by the esterases found in red blood cells' cytosol, as opposed to plasma cholinesterases or the acetylcholinesterase on red blood cell membranes.

Uses :

- It is used in treatment of Atrial fibrillation, Atrial flutter and Heart rhythm disorders.

Adverse effects :

- Low blood sugar
- Irritability
- Dizziness
- Nausea
- Fast heart rate
- Anxiety
- Coldness in hands and feet
- Pain or swelling near the injection site
- Weak and shallow breathing

Metoprolol :

Structure :



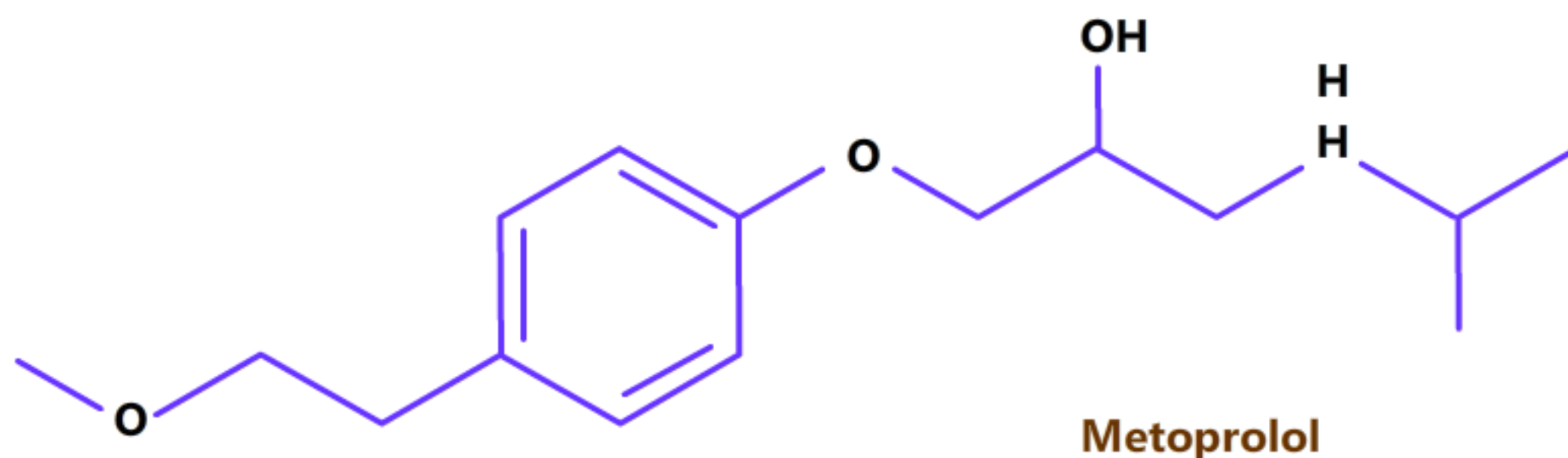


Fig 29: Structure of Metoprolol

- **IUPAC Name :** (RS)-1-[4-(2-Methoxyethyl)phenoxy]-3-[(propan2yl)amino]propan-2-ol

Properties :

- Physical appearance : Solid
- Melting point : 120°C
- Molecular weight : 237.36 g/mol
- Solubility : In Water



Mechanism :

- For cardiac cells, metoprolol functions as a β_1 -adrenergic antagonist.
- By producing negative inotropic effects, there is a decrease in the cardiac output.

Metabolism :

- The main metabolites of metoprolol by CYP2D6 are -hydroxymetoprolol and O-demethylmetoprolol.

Uses :

- It is used for treatment of Angina, Hypertension and Heart failure.

Adverse effects :

- Depression
- Confusion
- Memory problems

- Dizziness
- Tiredness
- Itching or rashes in skin
- Diarrhoea
- Insomnia
- Very slow heartbeats
- Lightheadedness
- Shortness of breath
- Coldness in palm and feet

Labetolol :

Structure :

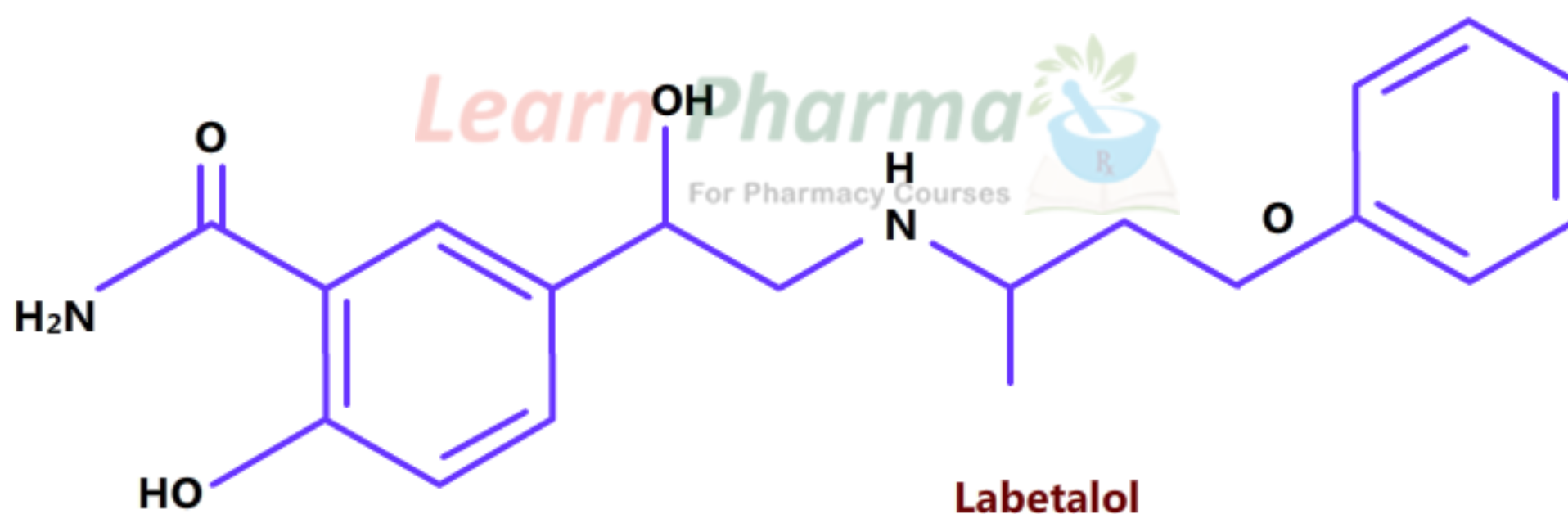


Fig 30: Structure of Labetalol

- **IUPAC Name :** (RS)-2-Hydroxy-5-[1-hydroxy-2-[(4-phenylbutan-2-yl)amino]ethyl]benzamide

Properties :

- Physical appearance : Solid
- Melting point : 188°C
- Solubility : In Water
- Molecular weight : 328.4 g/mol

Mechanism :

- It acts by relaxing blood vessels and slowing heart rate to improve blood flow and decrease blood pressure.

Metabolism :

- It is metabolised by the liver which results in an inactive glucuronide conjugate. It has an onset of action within 2 to 5 minutes, reaches its peak effects at 5 to 15 minutes, has an elimination half-life of 5.5 hours, and a duration of action up to four hours.

Uses :

- It is used for treatment of Prevention of strokes and Hypertension.

Adverse effects :

- Decreased libido
- Lightheadedness
- Dizziness
- Tiredness
- Scalp tingling



Carvedilol :

Structure :

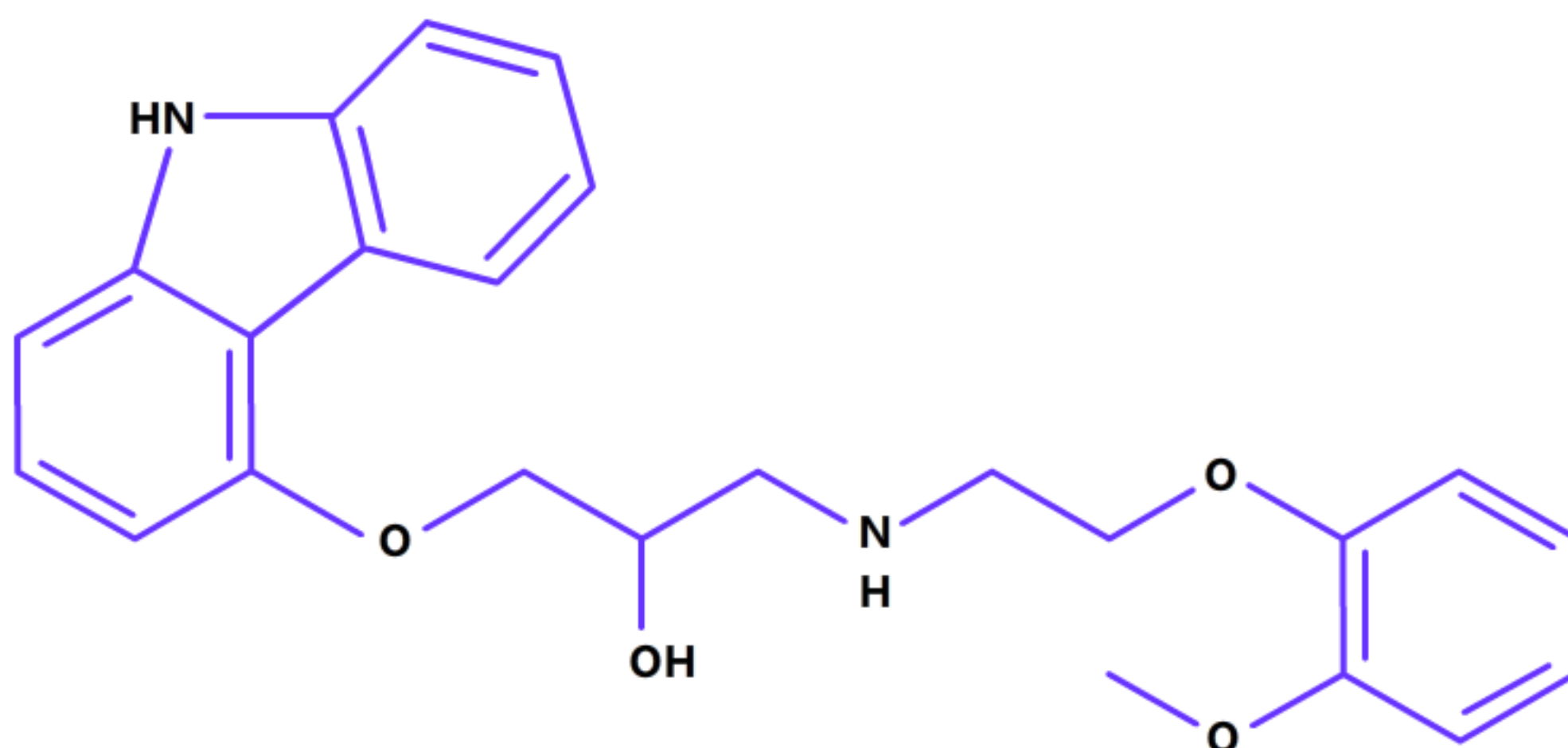


Fig 31: Structure of Carvedilol

- **IUPAC Name :** $(\pm)[3(9\text{Hcarbazol}4\text{yloxy})2\text{hydroxypropyl}][2(\text{methoxyphenoxy})\text{ethyamine}]$

Synthesis of Carvedilol :

- This was synthesised in two steps :
 - Condensation of 4-hydroxy carbazole with epichlorohydrin to produce 4-(2,3-epoxypropoxy)carbazole.
 - Which then was reacted with 2-(2-methoxyphenoxy)ethanamine (12.2. 30) to produce the desired carvedilol.

Properties :

- Physical appearance : Solid forms colourless crystals from ethyl acetate
- Melting point : 114.5°C
- Solubility : Practically insoluble in water; freely soluble in dimethylsulfoxide and soluble in methanol
- Molecular weight : 406.5



Mechanism :

- It produces its antihypertensive effect partly by reducing total peripheral resistance by blocking α 1-adrenoceptors and by preventing beta-adrenoceptor-mediated compensatory mechanisms.

Metabolism :

- It is metabolised primarily by aromatic ring oxidation and glucuronidation.
- The oxidative metabolites are then subjected to additional conjugation metabolism through glucuronidation and sulfation.
- The bile is the main route through which the carvedilol metabolites are eliminated into the faeces.

Uses :

- It is used to treat mild to severe chronic heart failure, hypertension, and left ventricular dysfunction in patients who are clinically stable after a myocardial infarction.

Adverse effects :

- ☐ Dizziness
- ☐ Tiredness
- ☐ Impotency
- ☐ Diarrhoea
- ☐ Drowsiness
- ☐ Lightheadedness

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